Time series regression and anomalies of errors

Econometrics. Lecture 10

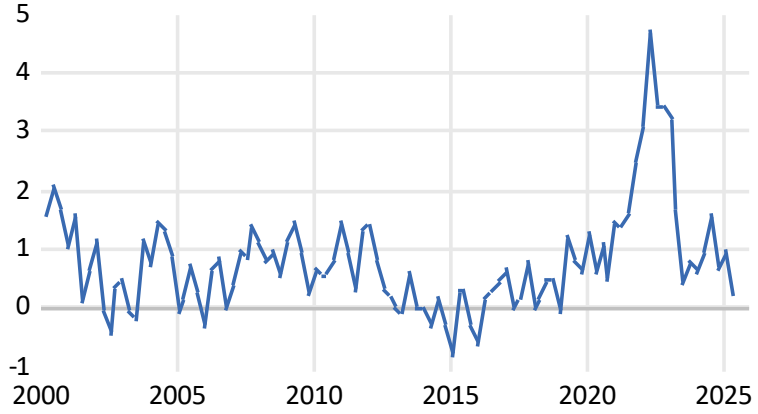
Content

1. Time series regression models
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4. Model properties when assumptions fails
5. Regression models diagnostics
6. Dealing with error irregularities

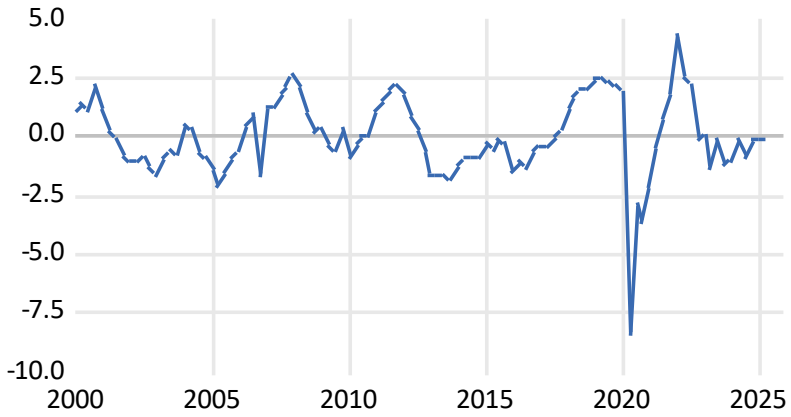
1. Time series regression models

Poland inflation (1)

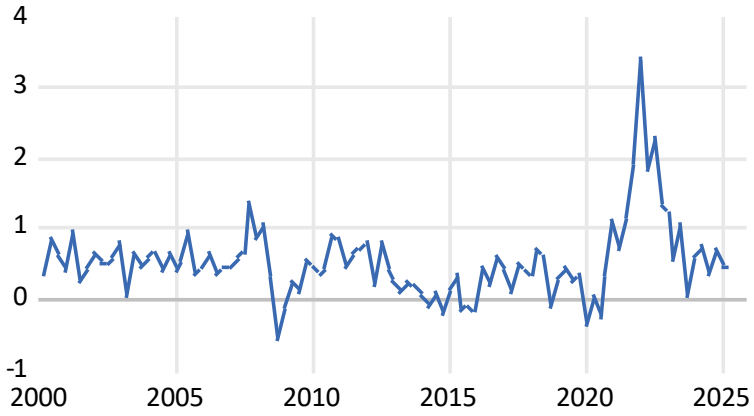
PL CPI Inflation, q-o-q, 100*log-diff



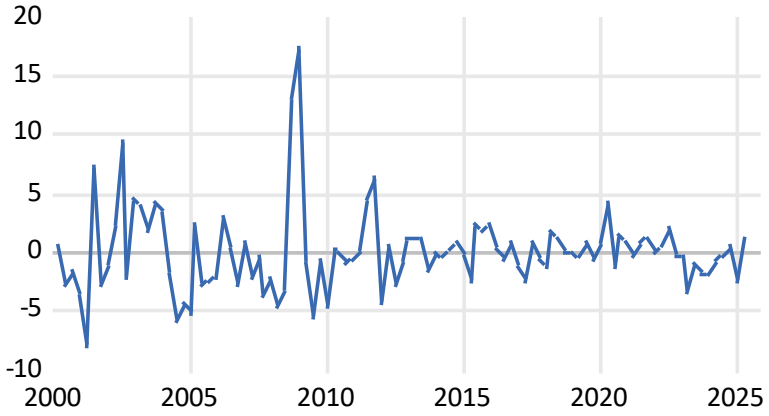
PL Output Gap (HP-filtred), %



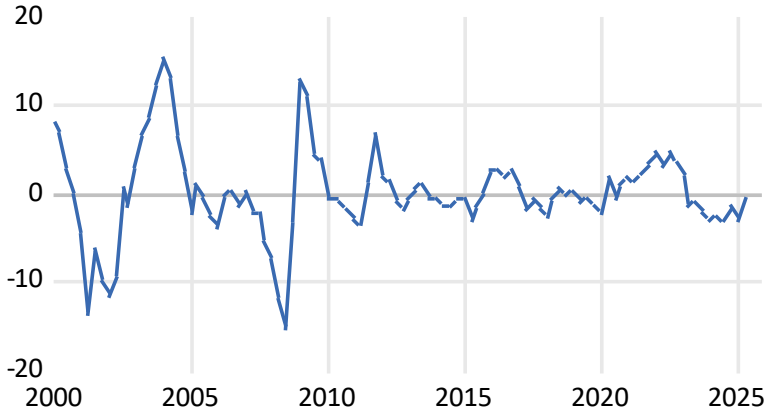
EA CPI Inflation, q-o-q, 100*log-diff



EUR/PLN, q-o-q, 100*log-diff



PL REER Gap (HP-filtred), %



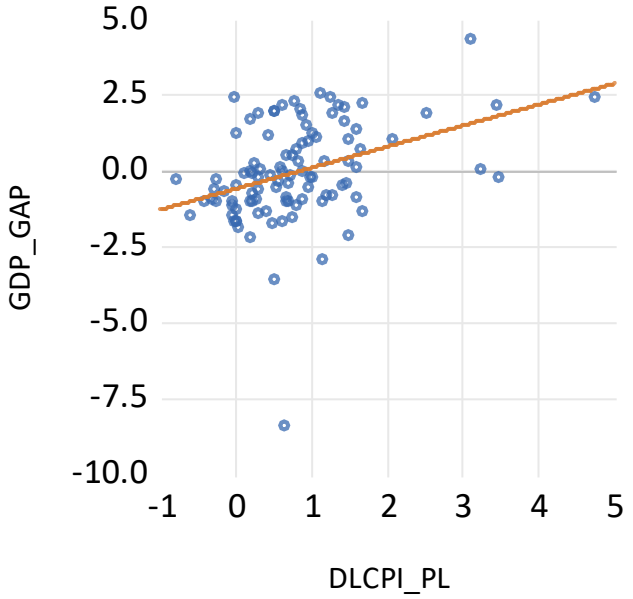
Static model

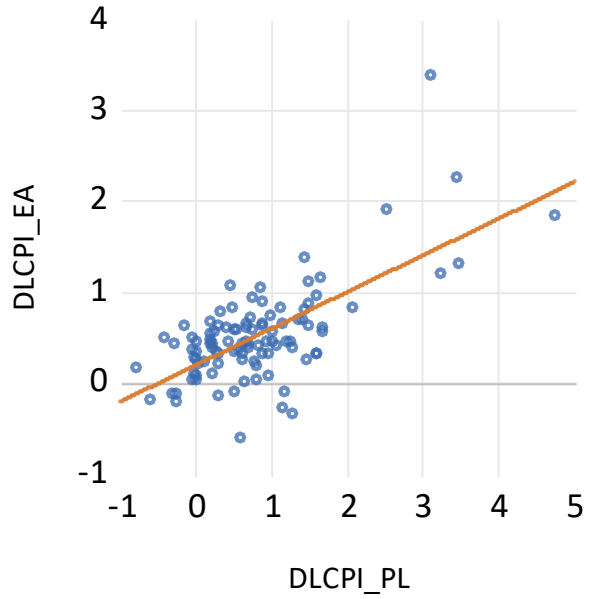
Determinants of inflation in Poland:

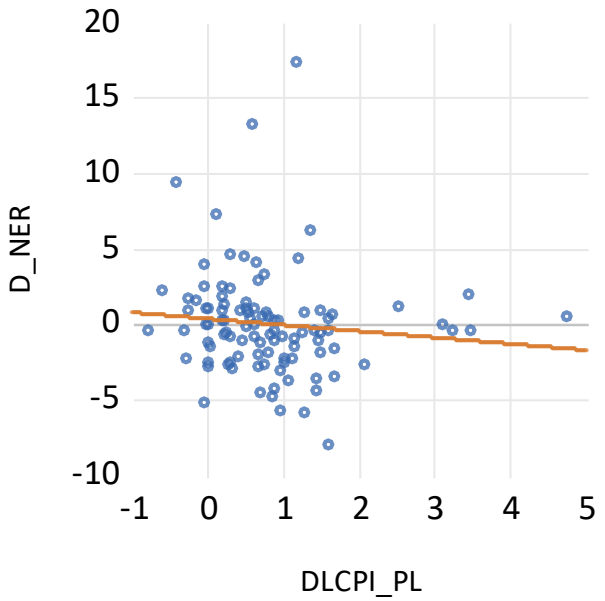
$$\Delta \ln cpi_t^{pl} = \beta_0 + \beta_1 \widehat{gdp}_t + \beta_2 \widehat{rer}_t^{eurpln} + \beta_3 \Delta \ln cpi_t^{ea} + \beta_4 \Delta \ln xr_t^{eurpln} + u_t$$

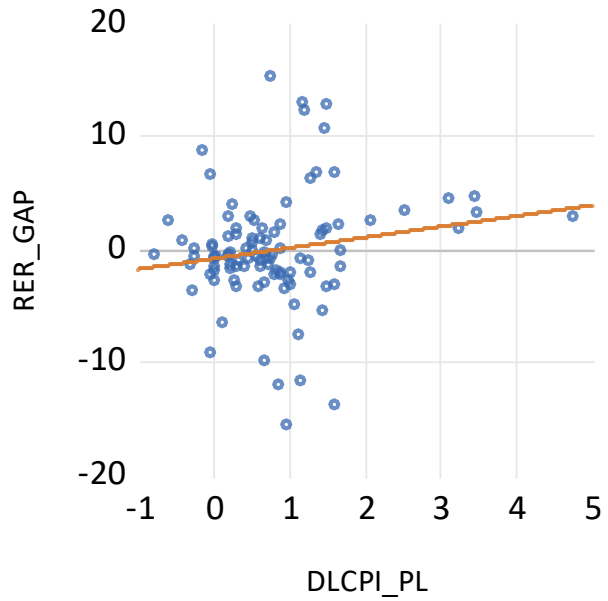
A **static model** is postulated when changes in explanatory variables (z) at time t are believed to have an immediate effect on y: $\Delta y_t = \beta \Delta z_t$, when $\Delta u_t = 0$.

Time series must be stationary or cointegrated!!!









Inflation in Poland. Model 1: Static

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:09
Sample (adjusted): 2000Q2 2025Q2
Included observations: 101 after adjustments

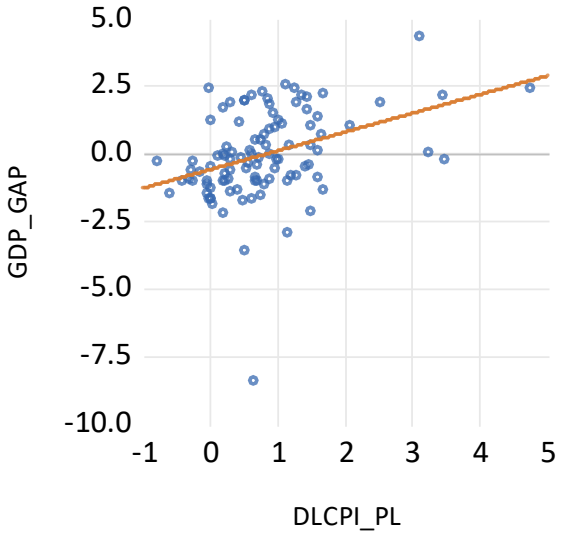
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.267581	0.098648	2.712479	0.0079
DLOG(EU_CPI_OBS)*100	1.009735	0.138825	7.273429	0.0000
DLOG(S_EUR_OBS)*100	-0.012042	0.020627	-0.583801	0.5607
GDP_GAP	0.071128	0.043616	1.630767	0.1062
RER_GAP	0.026761	0.014328	1.867733	0.0648

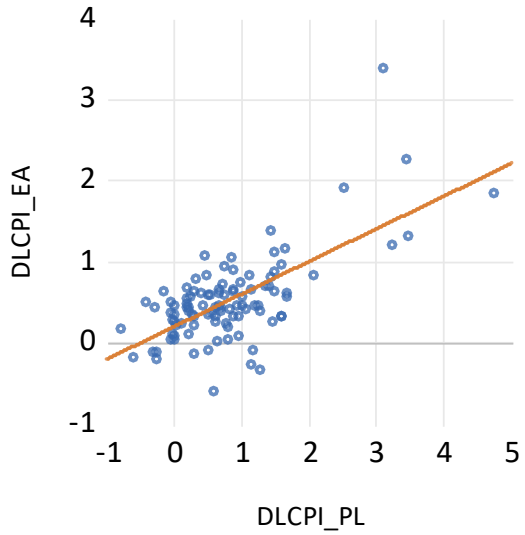
R-squared	0.484456	Mean dependent var	0.799165
Adjusted R-squared	0.462975	S.D. dependent var	0.892248
S.E. of regression	0.653857	Akaike info criterion	2.036382
Sum squared resid	41.04280	Schwarz criterion	2.165843
Log likelihood	-97.83729	Hannan-Quinn criter.	2.088792
F-statistic	22.55274	Durbin-Watson stat	1.358476
Prob(F-statistic)	0.000000		

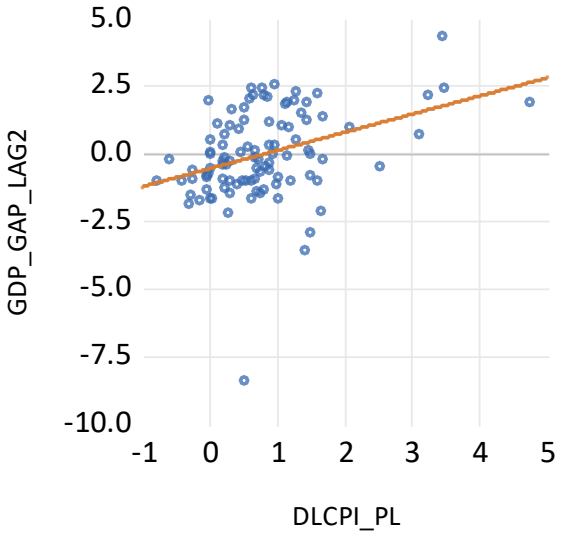
It's highly possible that our model is misspecified.

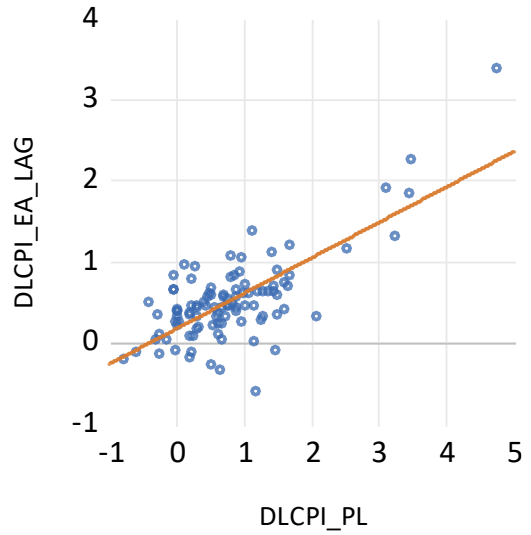
We don't account for the fact, that in reality changes in economic conditions influence on the behavior of economic agents with some time lag.

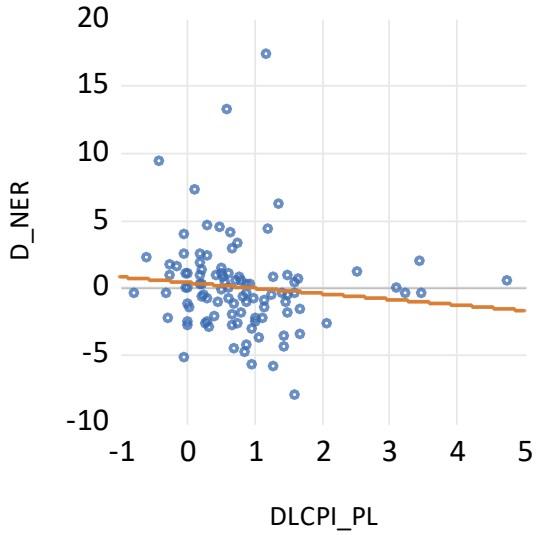
Time lags

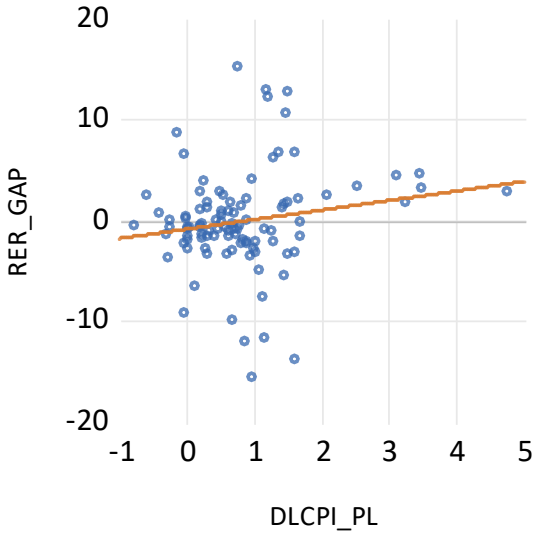

GDP_GAP
DLCPI_PL

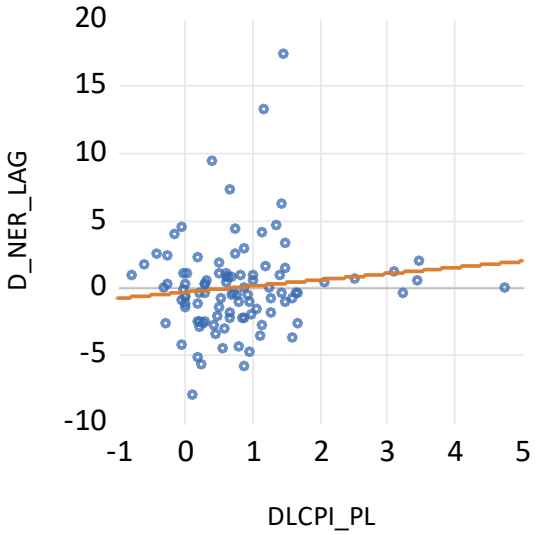

DLCPI_EA
DLCPI_PL

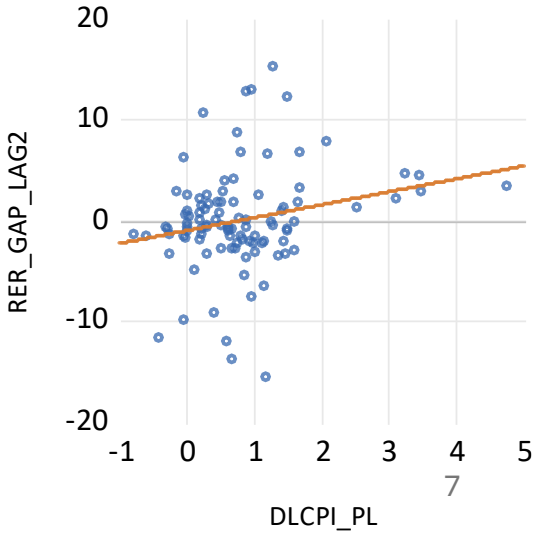

GDP_GAP_LAG2
DLCPI_PL


DLCPI_EA_LAG
DLCPI_PL


D_NER
DLCPI_PL


RER_GAP
DLCPI_PL


D_NER_LAG
DLCPI_PL


RER_GAP_LAG2
DLCPI_PL

Finite distributed lag models

In a **finite distributed lag (FDL) model**, we allow one or more variables to affect *y* with a lag.

$$\Delta \ln cpi_t^{pl} = \alpha_0 + \delta_1 \widehat{gdp}_{t-2} + \delta_2 \widehat{rer}_{t-1}^{eurpln} + \\ + \delta_3 \Delta \ln cpi_t^{ea} + \delta_4 \Delta \ln cpi_{t-1}^{ea} + \delta_5 \Delta \ln xr_{t-1}^{eurpln} + u_t$$

δ_3 is the immediate change in Poland inflation due to the 1 p.p. increase in EA inflation at time t. δ_3 is usually called the **impact multiplier**.

δ_4 is the change in Poland inflation in one period after the **temporary change** of EA inflation.

$\delta_3 + \delta_4$ is the change in Poland inflation in one period after the **permanent change** of EA inflation. The sum is called the **long-run multiplier**.

Inflation in Poland. Model 2: FDL

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:09
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

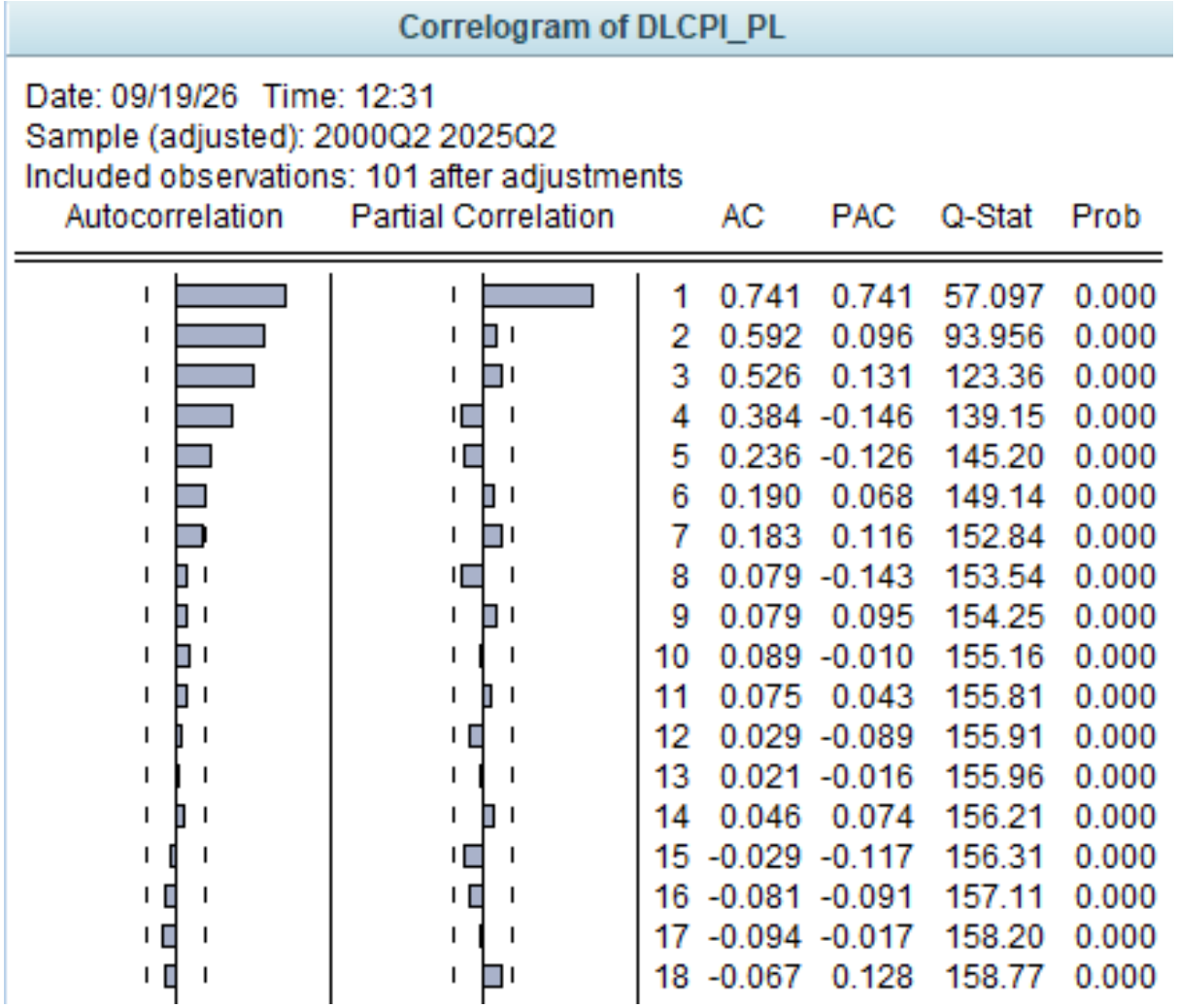
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.076033	0.076634	0.992159	0.3237
DLOG(EU_CPI_OBS)*100	0.526008	0.122728	4.285982	0.0000
DLOG(EU_CPI_OBS(-1))*100	0.822268	0.127554	6.446423	0.0000
DLOG(S_EUR_OBS(-1))*100	0.040880	0.015652	2.611813	0.0105
GDP_GAP(-2)	0.103230	0.031577	3.269132	0.0015
RER_GAP(-1)	0.029243	0.011005	2.657226	0.0093

R-squared	0.708101	Mean dependent var	0.791340
Adjusted R-squared	0.692575	S.D. dependent var	0.893252
S.E. of regression	0.495272	Akaike info criterion	1.490705
Sum squared resid	23.05767	Schwarz criterion	1.647016
Log likelihood	-68.53527	Hannan-Quinn criter.	1.553967
F-statistic	45.60594	Durbin-Watson stat	1.112405
Prob(F-statistic)	0.000000		

Coefficients on the explanatory variables are significant at 5% level. But the model is still misspecified.

We don’t account for the **persistence** in the inflation dynamics.

Inflation persistence



Correlogram of DLCPI_PL

Date: 09/19/26 Time: 12:31
Sample (adjusted): 2000Q2 2025Q2
Included observations: 101 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.741	0.741	57.097	0.000
		2	0.592	0.096	93.956	0.000
		3	0.526	0.131	123.36	0.000
		4	0.384	-0.146	139.15	0.000
		5	0.236	-0.126	145.20	0.000
		6	0.190	0.068	149.14	0.000
		7	0.183	0.116	152.84	0.000
		8	0.079	-0.143	153.54	0.000
		9	0.079	0.095	154.25	0.000
		10	0.089	-0.010	155.16	0.000
		11	0.075	0.043	155.81	0.000
		12	0.029	-0.089	155.91	0.000
		13	0.021	-0.016	155.96	0.000
		14	0.046	0.074	156.21	0.000
		15	-0.029	-0.117	156.31	0.000
		16	-0.081	-0.091	157.11	0.000
		17	-0.094	-0.017	158.20	0.000
		18	-0.067	0.128	158.77	0.000

Autoregressive distributed lag models

In an **autoregressive distributed lag (ADL) model**, we include lags of *y* as explanatory variables.

$$\Delta \ln cpi_t^{pl} = \mu_0 + \gamma_1 \Delta \ln cpi_{t-1}^{pl} + \gamma_2 \widehat{gdp}_{t-2} + \gamma_3 \widehat{rer}_{t-1}^{eurpln} + \\ + \gamma_4 \Delta \ln cpi_t^{ea} + \gamma_5 \Delta \ln cpi_{t-1}^{ea} + \gamma_6 \Delta \ln xr_{t-1}^{eurpln} + u_t$$

γ_1 measures the **persistence** of inflation in Poland.

γ_6 is the **short-run effect** of the exchange rate change on inflation in Poland (short-run exchange rate pass-through).

$\frac{\gamma_6}{1-\gamma_1}$ is the **long-run effect** of the exchange rate change on inflation in Poland (long-run exchange rate pass-through).

Inflation in Poland. Model 3: ADL

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:15
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000209	0.071828	0.002911	0.9977
DLOG(CPI_OBS(-1))*100	0.337564	0.075042	4.498336	0.0000
DLOG(EU_CPI_OBS)*100	0.415841	0.114470	3.632762	0.0005
DLOG(EU_CPI_OBS(-1))*100	0.562013	0.129821	4.329120	0.0000
DLOG(S_EUR_OBS(-1))*100	0.045946	0.014305	3.211898	0.0018
GDP_GAP(-2)	0.059109	0.030396	1.944599	0.0548
RER_GAP(-1)	0.020443	0.010216	2.001096	0.0483

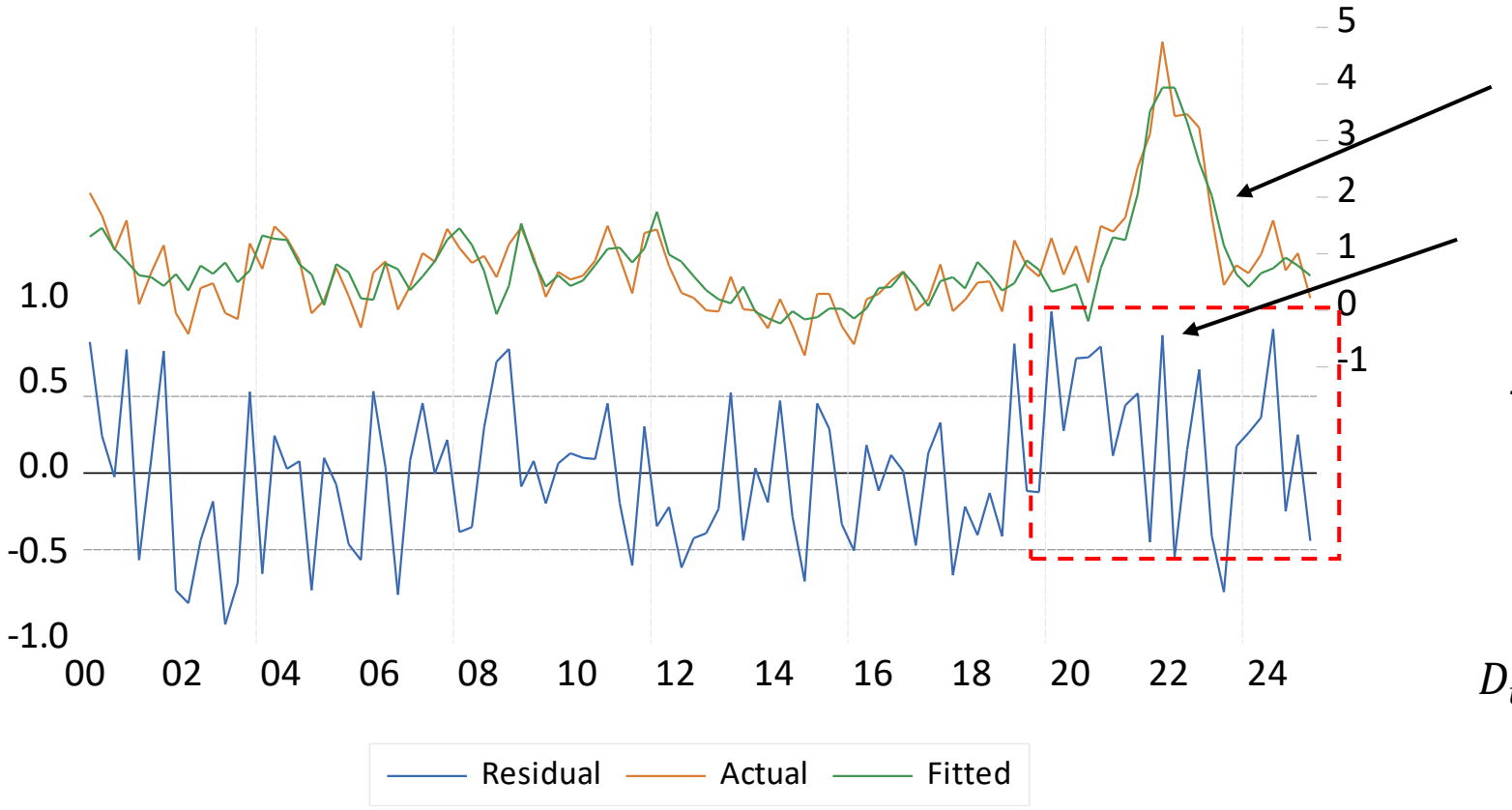
R-squared	0.760264	Mean dependent var	0.791340
Adjusted R-squared	0.744797	S.D. dependent var	0.893252
S.E. of regression	0.451250	Akaike info criterion	1.313839
Sum squared resid	18.93728	Schwarz criterion	1.496201
Log likelihood	-58.69197	Hannan-Quinn criter.	1.387645
F-statistic	49.15433	Durbin-Watson stat	1.868432
Prob(F-statistic)	0.000000		

The results seem economically reasonable.

But we must analyse the residuals to make a conclusion about the model adequacy.

2. Dummy variables and structural breaks

Residuals from Model 3



Inflation is higher in recent years.

Residuals are mostly positive and correlated after 2020 – an indicator of structural break at an intercept.

To account for these break, we can include the dummy variable into the model:

$$D_t^{2020s} = \begin{cases} 1 & \text{for } t = \text{Q1 2020} - \text{Q4 2025} \\ 0 & \text{otherwise} \end{cases}$$

$$\Delta \ln cpi_t^{pl} = \mu_0 + \gamma_1 \Delta \ln cpi_{t-1}^{pl} + \gamma_2 \widehat{gdp}_{t-2} + \gamma_3 \widehat{rer}_{t-1}^{eurpln} + \\ + \gamma_4 \Delta \ln cpi_t^{ea} + \gamma_5 \Delta \ln cpi_{t-1}^{ea} + \gamma_6 \Delta \ln xr_{t-1}^{eurpln} + \gamma_7 D_t^{2020s} + u_t$$

Inflation in Poland.
Model 4: Including a level-shift dummy

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 08/03/26 Time: 18:36
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.027518	0.067990	0.404741	0.6866
DLOG(CPI_OBS(-1))*100	0.198519	0.080375	2.469903	0.0154
DLOG(EU_CPI_OBS)*100	0.378448	0.108178	3.498384	0.0007
DLOG(EU_CPI_OBS(-1))*100	0.564788	0.122127	4.624579	0.0000
GDP_GAP(-2)	0.098296	0.030577	3.214721	0.0018
RER_GAP(-1)	0.025991	0.009732	2.670685	0.0090
DLOG(S_EUR_OBS(-1))*100	0.041422	0.013515	3.064867	0.0029
D2020S	0.470506	0.130039	3.618186	0.0005

R-squared	0.790128	Mean dependent var	0.791340
Adjusted R-squared	0.774159	S.D. dependent var	0.893252
S.E. of regression	0.424498	Akaike info criterion	1.200799
Sum squared resid	16.57825	Schwarz criterion	1.409212
Log likelihood	-52.03994	Hannan-Quinn criter.	1.285147
F-statistic	49.48024	Durbin-Watson stat	1.794810
Prob(F-statistic)	0.000000		

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:15
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

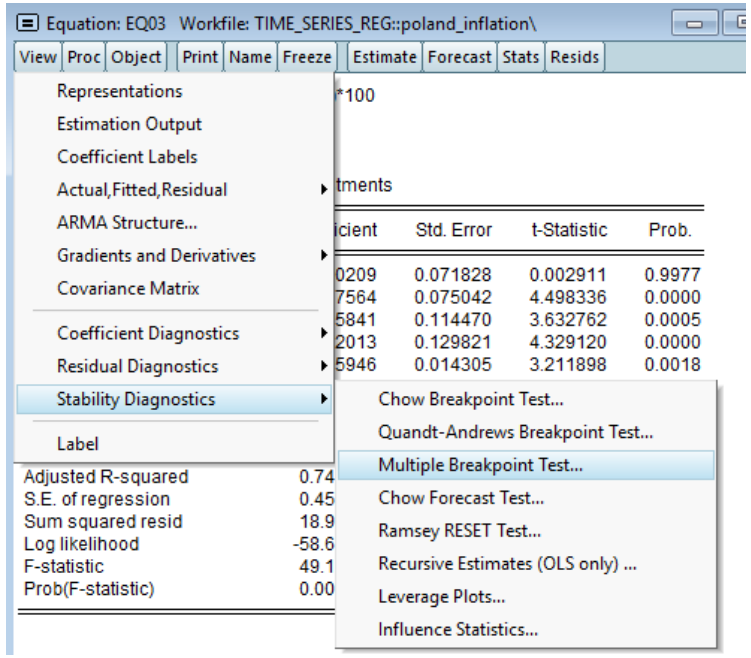
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000209	0.071828	0.002911	0.9977
DLOG(CPI_OBS(-1))*100	0.337564	0.075042	4.498336	0.0000
DLOG(EU_CPI_OBS)*100	0.415841	0.114470	3.632762	0.0005
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GDP_GAP(-2)	0.059109	0.030396	1.944599	0.0548
RER_GAP(-1)	0.020443	0.010216	2.001096	0.0483

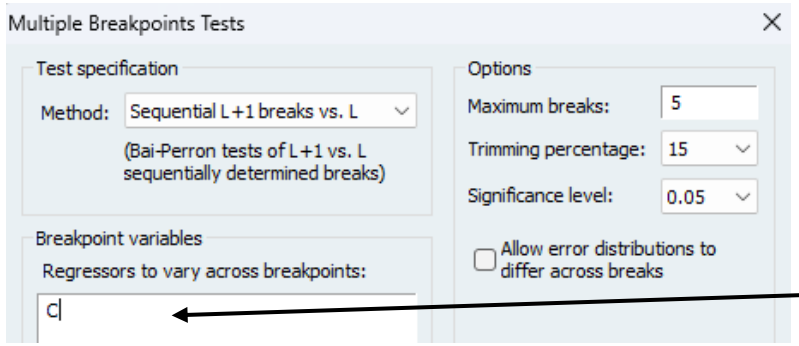
R-squared	0.760264	Mean dependent var	0.791340
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S.E. of regression	0.451250	Akaike info criterion	1.313839
Sum squared resid	18.93728	Schwarz criterion	1.496201
Log likelihood	-58.69197	Hannan-Quinn criter.	1.387645
F-statistic	49.15433	Durbin-Watson stat	1.868432
Prob(F-statistic)	0.000000		

During each quarter of 2020-2025, inflation in Poland was on average higher by about 0.47 p.p. than in other periods, conditional on the factors included in the model.

Structural break in the intercept. Bai-Perron test

The Bai-Perron test checks whether a regression's coefficients have shifted at some unknown point (or points) in the sample, rather than staying fixed throughout.





Specify which variables to test for breaks (in our case only the intercept)

Multiple breakpoint tests
Bai-Perron tests of L+1 vs. L sequentially determined breaks
Date: 08/03/26 Time: 19:12
Sample: 2000Q1 2025Q4
Included observations: 100
Breaking variables: C
Non-breaking variables: DLOG(CPI_OBS(-1))*100
DLOG(EU_CPI_OBS)*100 DLOG(EU_CPI_OBS(-1))*100
DLOG(S_EUR_OBS(-1))*100 GDP_GAP(-2) RER_GAP(-1)
Break test options: Trimming 0.15, Max. breaks 5, Sig. level 0.05

Sequential F-statistic determined breaks: 1

Break Test	F-statistic	Scaled F-statistic	Critical Value**
0 vs. 1 *	13.09127	13.09127	8.58
1 vs. 2	3.317385	3.317385	10.13

* Significant at the 0.05 level.
** Bai-Perron (Econometric Journal, 2003) critical values.

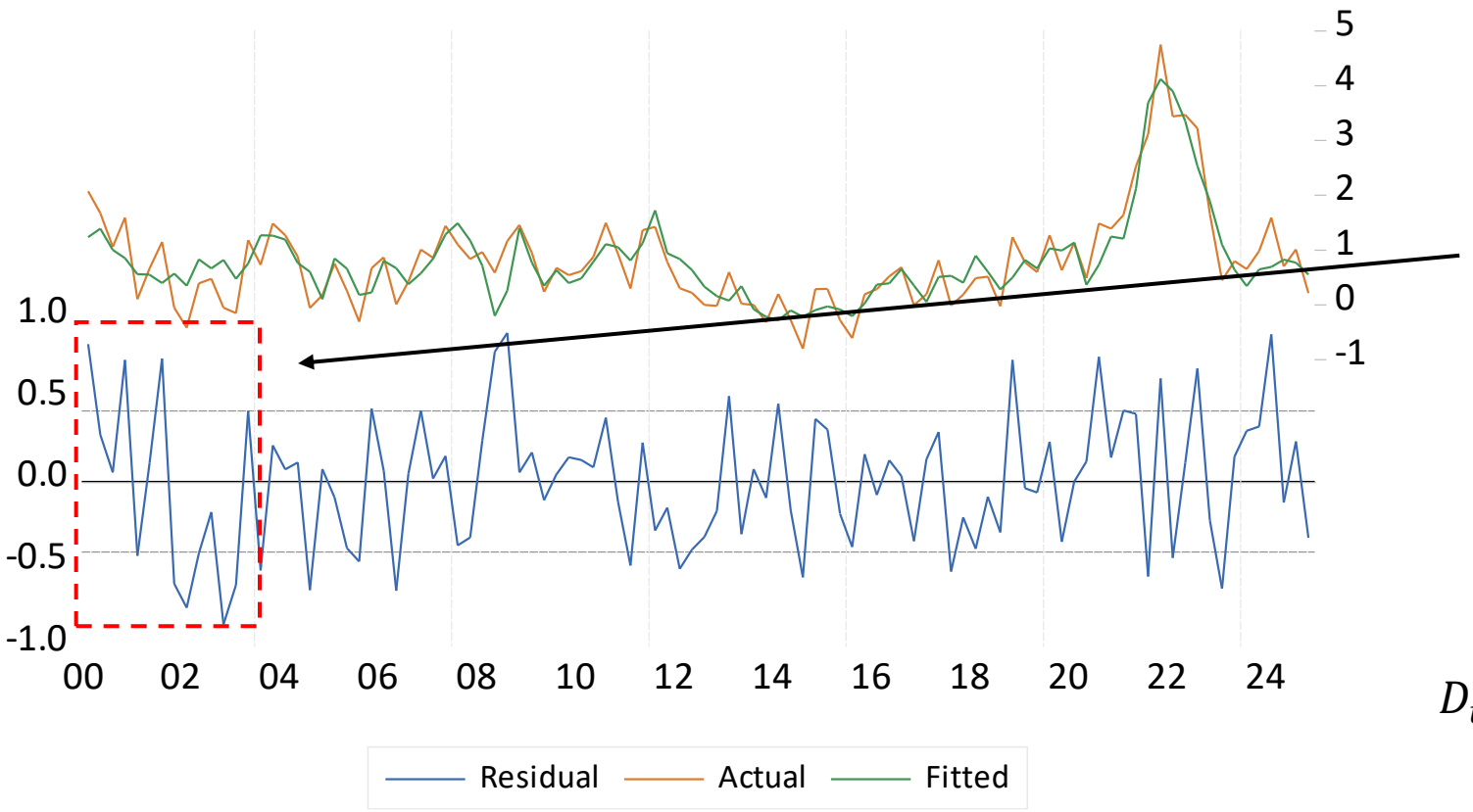
Break dates:

	Sequential	Repartition
1	2020Q1	2020Q1

The date of the structural break

The test works sequentially: it first asks whether allowing one break improves the fit enough to justify the extra parameters (testing 0 vs. 1 break), and if so, keeps testing whether one more break (1 vs. 2, and so on) adds significant explanatory power, stopping as soon as an additional break no longer clears the bar.

Residuals from Model 4



In 2000-2001 residuals were always positive and probably correlated, and in 2002-2003 – negative and probably correlated.

It's highly possible that during the transition period (2000–2003) inflation may have responded to the state of aggregate demand differently.

$$D_t^{2003} = \begin{cases} 1 & \text{for } t = \text{Q1 2000} - \text{Q4 2003} \\ 0 & \text{otherwise} \end{cases}$$

$$\Delta \ln cpi_t^{pl} = \mu_0 + \gamma_1 \Delta \ln cpi_{t-1}^{pl} + \gamma_2 \widehat{gdp}_{t-2} + \gamma_3 \widehat{rer}_{t-1}^{eurpln} + \\ + \gamma_4 \Delta \ln cpi_t^{ea} + \gamma_5 \Delta \ln cpi_{t-1}^{ea} + \gamma_6 \Delta \ln xr_{t-1}^{eurpln} + \gamma_7 D_t^{2020s} + \gamma_8 D_t^{2003} \widehat{gdp}_{t-2} + u_t$$

Inflation in Poland.
Model 5: Including a slope-shift dummy

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 08/03/26 Time: 18:37
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.034448	0.062867	0.547945	0.5851
DLOG(CPI_OBS(-1))*100	0.116344	0.076967	1.511606	0.1341
DLOG(EU_CPI_OBS)*100	0.364470	0.100048	3.642941	0.0004
DLOG(EU_CPI_OBS(-1))*100	0.680303	0.116372	5.845943	0.0000
GDP_GAP(-2)	0.077938	0.028698	2.715748	0.0079
RER_GAP(-1)	0.026603	0.008997	2.957078	0.0040
DLOG(S_EUR_OBS(-1))*100	0.056780	0.013046	4.352418	0.0000
D2020S	0.508820	0.120562	4.220407	0.0001
D2003*GDP_GAP(-2)	0.407887	0.099857	4.084710	0.0001

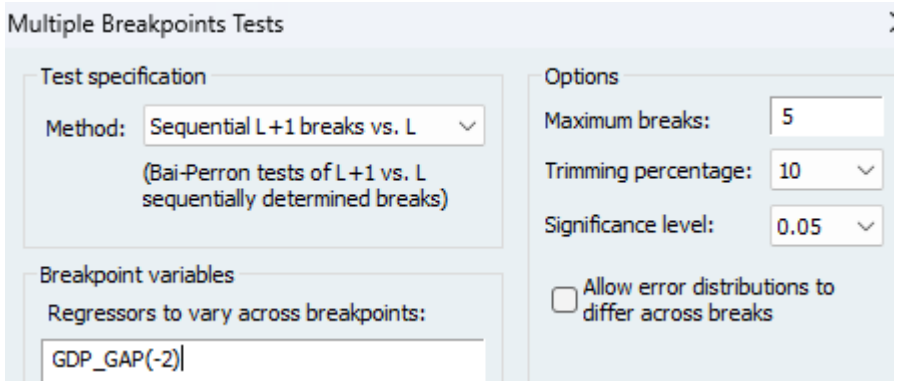
R-squared	0.822646	Mean dependent var	0.791340
Adjusted R-squared	0.807054	S.D. dependent var	0.893252
S.E. of regression	0.392367	Akaike info criterion	1.052449
Sum squared resid	14.00959	Schwarz criterion	1.286915
Log likelihood	-43.62247	Hannan-Quinn criter.	1.147342
F-statistic	52.76211	Durbin-Watson stat	1.871689
Prob(F-statistic)	0.000000		

Outside 2000-2003, the short-term impact of the 1 p.p. change in the output gap on inflation was 0.078 p.p. (after two quarters).

During 2000-2003, the short-term impact of the output gap on inflation was 0.41 p.p. stronger than in later periods:

$$\gamma_2 + \gamma_8 = 0.077938 + 0.407887 \approx 0.486$$

Structural break in the slope. Bai-Perron test



Multiple breakpoint tests
Bai-Perron tests of L+1 vs. L sequentially determined breaks
Date: 08/03/26 Time: 19:33
Sample: 2000Q1 2025Q4
Included observations: 100
Breaking variables: GDP GAP(-2)
Non-breaking variables: C DLOG(CPI_OBS(-1))*100 DLOG(EU CPI OBS)*100 DLOG(EU CPI OBS(-1))*100 RER_GAP(-1) DLOG(S_EUR_OBS(-1))*100 D2020S
Break test options: Trimming 0.10, Max. breaks 5, Sig. level 0.05

Sequential F-statistic determined breaks: 1

Break Test	F-statistic	Scaled F-statistic	Critical Value**
0 vs. 1 *	20.10436	20.10436	9.10
1 vs. 2	3.581392	3.581392	10.55

* Significant at the 0.05 level.
** Bai-Perron (Econometric Journal, 2003) critical values.

Break dates:

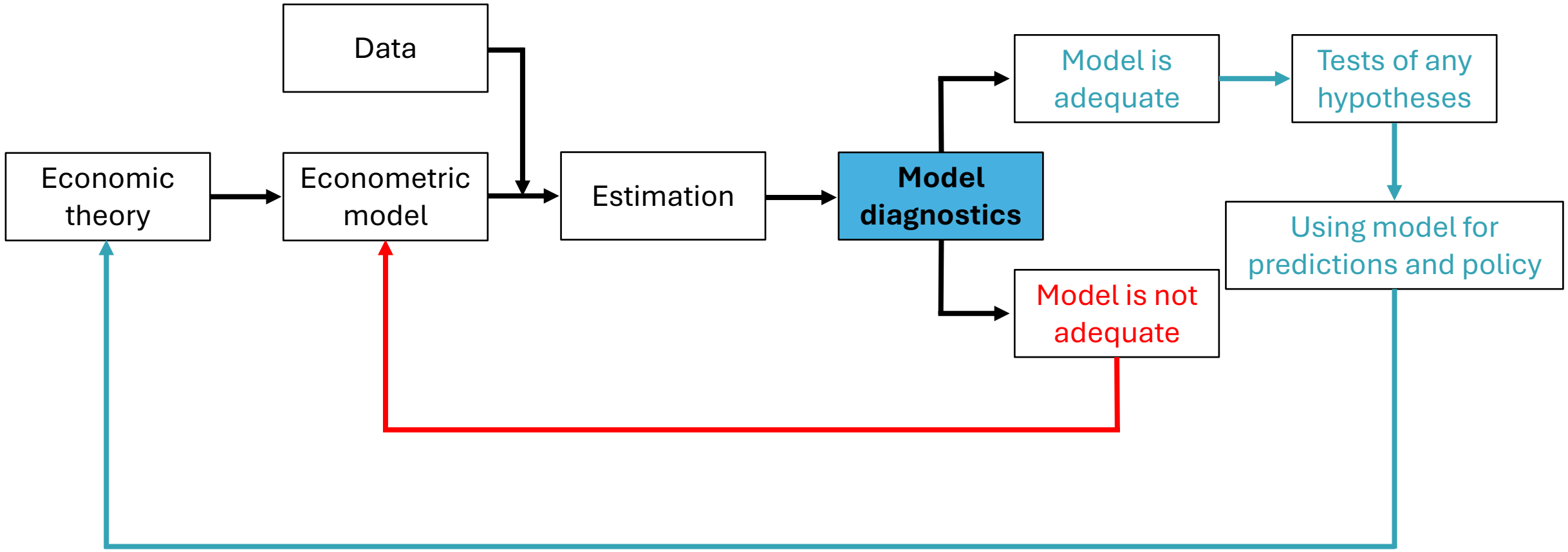
	Sequential	Repartition
1	2003Q4	2003Q4

Structural breaks may cause:

- Worse forecasting performance in terms of the mean forecast.
- Interval forecasts are wrong in both directions at once.

3. OLS assumptions for time series regression

Stages of econometric analysis



Linear regression model

Linear regression:

$$y_t = \beta_0 + \beta_1 x_{1t} + \cdots + \beta_k x_{kt} + u_t$$

In a matrix form

$$Y = X\beta + u$$

$$Y = (y_1, \dots, y_T)'$$

$$\beta = (\beta_0, \beta_1, \dots, \beta_k)'$$

$$u = (u_1, \dots, u_T)'$$

$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{k1} \\ \dots & \dots & \dots & \dots \\ 1 & x_{1T} & \dots & x_{kT} \end{pmatrix}$$

OLS assumptions TS.1 – TS.3

Assumption TS.1 (Linear in Parameters)

- Linear relationship between *Y* and *X*

Assumption TS.2 (No Perfect Collinearity)

- In the sample (and therefore in the underlying time series process), no independent variable is constant nor a perfect linear combination of the others.

Assumption TS.3 (Zero Conditional Mean)

- For each t, the expected value of the error u_t, given the explanatory variables for *all* time periods, is zero: $\mathrm{E}(u_t|X) = 0, t = 1,2, \dots, n.$

When TS.3 holds, we say that the explanatory variables are **strictly exogenous.**

Under Assumptions TS.1, TS.2, and TS.3, the OLS estimators are **unbiased** conditional on *X*, and therefore unconditionally as well.

Weak exogeneity

Strict exogeneity can be unrealistic.

Contemporaneous exogeneity is sufficient for proving *consistency* of the OLS estimator:

$$\mathrm{E}(u_t|x_{t1}, \ldots, x_{tk}) = \mathrm{E}(u_t|x_t) = 0$$

When u_t and the explanatory variables are contemporaneously uncorrelated, we say that the explanatory variables are **weakly exogenous.**

OLS assumptions TS.4 – TS.6

Assumption TS.4 (Homoskedasticity)

- Conditional on *X*, the variance of u_t is the same for all t:

$$\mathrm{Var}(u_t|X) = \mathrm{Var}(u_t) = \sigma^2, t = 1,2, \dots, n.$$

Assumption TS.5 (No Serial Correlation)

- Conditional on *X*, the errors in two different time periods are uncorrelated:

$$\mathrm{Corr}(u_t, u_s|X) = 0, \text{for } t \neq s.$$

Under Assumptions TS.1 – TS.5, the OLS estimators are **the best linear unbiased estimators** conditional on *X*.

Assumption TS.6 (Normality)

- The errors u_t are independent of *X* and are independently and identically distributed as Normal(0, σ^2).

What each assumption buys you

- TS.1 to TS.3 give **unbiasedness**.
- Add TS.4 and TS.5 and OLS is **efficient** among linear unbiased estimators, and the conventional variance formula is the right one.
- Add TS.6 and inference is **exact in finite samples**: *t* and *F* statistics have their textbook distributions at any sample size.
- Drop TS.6 alone and little is lost. The statistics remain usable asymptotically, and the normal and χ^2 approximations are usually adequate at macro sample sizes. This is the least important assumption on the list, which is the opposite of how much attention it usually gets.

4. Model properties when assumptions fail

Classical linear model assumptions for time series regression

Assumption TS.1 (Linear in parameters)	$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$
Assumption TS.2 (No perfect collinearity)	None of the independent variables is constant, and there are no exact linear relationships among them
Assumption TS.3 (Zero conditional mean)	$\mathrm{E}(u_t \mid X) = 0, t = 1,2, \ldots, n$
Assumption TS.4 (Homoskedasticity)	$\mathrm{Var}(u_t \mid X) = \mathrm{Var}(u_t) = \sigma^2, t = 1,2, \ldots, n.$
Assumption TS.5 (No serial correlation)	$\mathrm{Corr}(u_t, u_s \mid X) = 0, \text{for } t \neq s.$
Assumption TS.6 (Normality)	$u_t \sim \mathrm{Normal}(0, \sigma^2).$

Note:

- TS.1 and TS.2 are rarely the problem. Linearity in parameters is a property of how you wrote the model, and perfect collinearity announces itself by making the model unestimable.

When a regressor is correlated with the error

Example (two regressors – x_t, z_t: $E(z_t u_t) \neq 0$):

$$y_t = \beta_1 x_t + \beta_2 z_t + u_t$$

- Both coefficients are biased and inconsistent.
- The exception: $\hat{\beta}_1$ escapes only if x_t is uncorrelated with both z_t and y_t.
- The bias does not shrink with the sample.

When a regressor is correlated with the error. Example

Generated data:

y = 0.75*x + 0.60*z + u

x = 0.7*x(-1) + 2*nrnd

z = 0.5*z(-1) + 1.5*u+nrnd

u = nrnd

Estimated model:

y = c(1)*x + c(2)*z

True DGP

Dependent Variable: Y
Method: Least Squares
Date: 05/25/26 Time: 16:51
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
X	0.750000	4.06E-18	1.85E+17	0.0000
Z	0.600000	8.57E-18	7.00E+16	0.0000
U	1.000000	1.70E-17	5.89E+16	0.0000

R-squared	1.000000	Mean dependent var	0.110392
Adjusted R-squared	1.000000	S.D. dependent var	3.024255
S.E. of regression	3.89E-16	Sum squared resid	1.50E-28
Durbin-Watson stat	1.485691		

Estimated model

Dependent Variable: Y
Method: Least Squares
Date: 05/25/26 Time: 16:52
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
X	0.738264	0.007570	97.52142	0.0000
Z	0.951017	0.011499	82.70675	0.0000

R-squared	0.942543	Mean dependent var	0.110392
Adjusted R-squared	0.942485	S.D. dependent var	3.024255
S.E. of regression	0.725284	Akaike info criterion	2.197494
Sum squared resid	524.4590	Schwarz criterion	2.207317
Log likelihood	-1095.648	Hannan-Quinn criter.	2.201227
Durbin-Watson stat	2.156964		

The coefficient *c(2)* is biased

Autocorrelated errors with exogenous regressors: $\mathrm{Corr}(u_t, u_{t-s}|X) \neq 0$, for some s

Example 1 (autocorrelated errors with exogenous regressors):

$$y_t = \beta_1 x_t + u_t$$

$$u_t = \rho u_{t-1} + v_t$$

- $\hat{\beta}_1$ remains unbiased and consistent.
- The reported standard error is wrong.
- Both the point forecast and the interval suffer.

Autocorrelated errors with exogenous regressors: $\mathrm{Corr}(u_t, u_{t-s}|X) \neq 0$, for some s. Example

Generated data:

y = 0.75*z + u

z = 0.9*z(-1) + 2*nrnd

u = 0.9*u(-1) + 1.2*nrnd

Estimated model:

y = c(1)*z

True DGP

Dependent Variable: Y
Method: Least Squares
Date: 02/01/26 Time: 14:26
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Z	0.750000	3.67E-18	2.05E+17	0.0000
U	1.000000	5.73E-18	1.75E+17	0.0000

R-squared	1.000000	Mean dependent var	-0.442822
Adjusted R-squared	1.000000	S.D. dependent var	4.366115
S.E. of regression	5.00E-16	Sum squared resid	2.50E-28
Durbin-Watson stat	0.661602		

Estimated model

Dependent Variable: Y
Method: Least Squares
Date: 02/01/26 Time: 14:26
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Z	0.787798	0.020227	38.94715	0.0000

R-squared	0.599076	Mean dependent var	-0.442822
Adjusted R-squared	0.599076	S.D. dependent var	4.366115
S.E. of regression	2.764562	Akaike info criterion	4.872642
Sum squared resid	7627.517	Schwarz criterion	4.877554
Log likelihood	-2432.885	Hannan-Quinn criter.	4.874509
Durbin-Watson stat	0.174732		

The coefficient *c(1)* is unbiased, … but the SE is invalid

Autocorrelated errors with a lagged dependent variable

Example 2 (autocorrelated errors with a lagged dependent variable):

$$y_t = \beta_1 x_t + \beta_2 y_{t-1} + u_t$$

$$u_t = \rho u_{t-1} + v_t$$

- The coefficients are biased and inconsistent:

  - y_{t-1} contains u_{t-1}, and u_{t-1} is correlated with u_t. So the regressor y_{t-1} is also correlated with the error u_t.

  This is TS.3 failing, arrived at through TS.5.

Autocorrelated errors with a lagged dependent variable. Example

Generated data:

y = 0.5*y(-1) + 0.75*z + u

z = 0.9*z(-1) + 2*nrnd

u = 0.9*u(-1) + 1.2*nrnd

Estimated model:

y = c(1)*y(-1) + c(2)*z

True DGP

Dependent Variable: Y
Method: Least Squares
Date: 02/01/26 Time: 14:29
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Y(-1)	0.500000	2.22E-17	2.25E+16	0.0000
Z	0.750000	3.40E-17	2.20E+16	0.0000
U	1.000000	4.83E-17	2.07E+16	0.0000

R-squared	1.000000	Mean dependent var	-0.873482
Adjusted R-squared	1.000000	S.D. dependent var	8.211234
S.E. of regression	2.73E-15	Sum squared resid	7.41E-27
Durbin-Watson stat	0.627532		

Estimated model

Dependent Variable: Y
Method: Least Squares
Date: 02/01/26 Time: 14:29
Sample: 2 1000
Included observations: 999

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Y(-1)	0.850462	0.009411	90.37210	0.0000
Z	0.330713	0.017942	18.43188	0.0000

R-squared	0.952593	Mean dependent var	-0.873482
Adjusted R-squared	0.952545	S.D. dependent var	8.211234
S.E. of regression	1.788747	Akaike info criterion	4.002907
Sum squared resid	3190.015	Schwarz criterion	4.012731
Log likelihood	-1997.452	Hannan-Quinn criter.	4.006641
Durbin-Watson stat	0.880341		

The coefficients *c(1)* and *c(2)* are biased

Errors are heteroskedastic

Example (errors variance σ_t^2 is time dependent):

$$y_t = \beta x_t + u_t$$

$$Var(u_t) = \sigma_t^2$$

- $\hat{\beta}$ is still unbiased:
    - but $\hat{\beta}$ is no longer efficient;
    - standard errors can no longer be used in inference (e.g., testing hypothesis).

Non-normal errors

Example:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$$

$$u_t \text{ is not} \sim N(0, \sigma^2).$$

- Small samples: *t* and *F* statistics no longer have their exact distributions.

- Large samples: OLS stays consistent, and the test statistics converge to known limiting distributions:

  - Inference is available

5. Regression models diagnostics

Portmanteau (Ljung–Box) Q test

- Regression model: $y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$

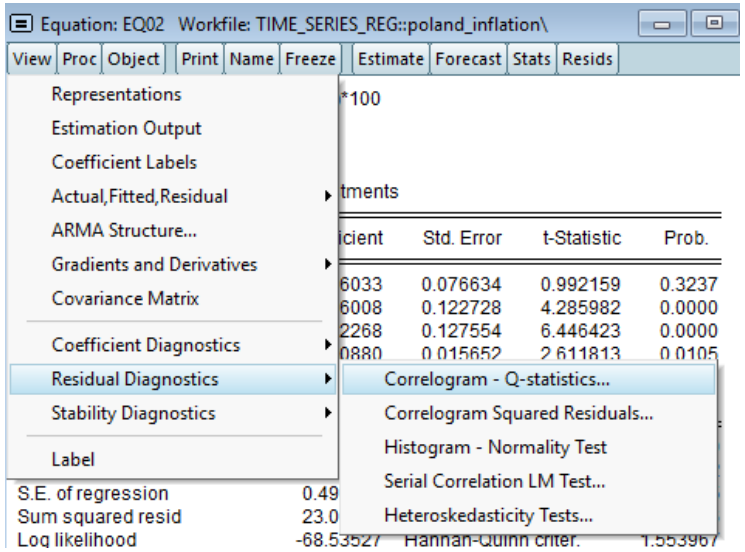
- Residuals: $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t} - \hat{\beta}_2 x_{2t} - \cdots - \hat{\beta}_k x_{kt}$

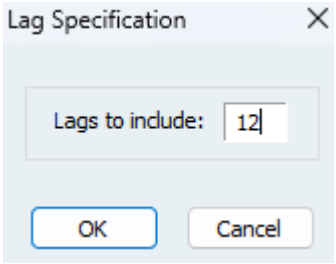
- Computes the residual autocorrelations at lags 1 to *m* and asks whether they are jointly zero. H_0: no autocorrelation up to order *m*.

- Choice of *m* is yours and it matters. For quarterly data, *m* of 4 and 8 are the natural reporting choices.

Portmanteau (Ljung–Box) Q test in EViews

Consider the estimated FDL model (2) for Poland inflation





Date: 02/04/26 Time: 16:53
Sample (adjusted): 2000Q3 2025Q2
Q-statistic probabilities adjusted for 5 dynamic regressors

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.423	0.423	18.405	0.000
		2	0.229	0.062	23.883	0.000
		3	0.293	0.215	32.932	0.000
		4	0.149	-0.063	35.277	0.000
		5	0.069	-0.014	35.790	0.000
		6	0.170	0.123	38.942	0.000
		7	0.088	-0.041	39.793	0.000
		8	-0.092	-0.168	40.732	0.000
		9	0.063	0.132	41.179	0.000
		10	0.027	-0.046	41.265	0.000
		11	-0.079	-0.040	41.979	0.000
		12	-0.011	-0.001	41.993	0.000

*Probabilities may not be valid for this equation specification.

p-values < 0.05 for all lags

We reject H_0: *no autocorrelation.*

Testing residuals for autocorrelation. Breusch-Godfrey LM test

- Regression model: $y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$

- Residuals: $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t} - \hat{\beta}_2 x_{2t} - \cdots - \hat{\beta}_k x_{kt}$

- Regress residuals $\hat{u}_t$ on x_{it} and lagged residuals $\hat{u}_{t-i}$:

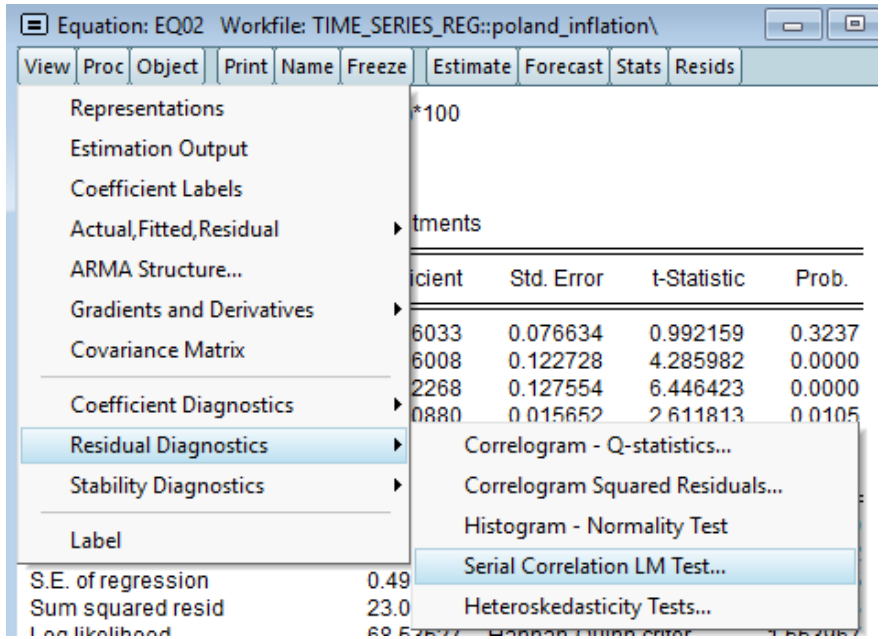
$$\hat{u}_t = \gamma_0 + \gamma_1 x_{1t} + \gamma_2 x_{2t} + \cdots + \gamma_k x_{kt} + \psi_1 \hat{u}_{t-1} + \cdots + \psi_m \hat{u}_{t-m} + v_t$$

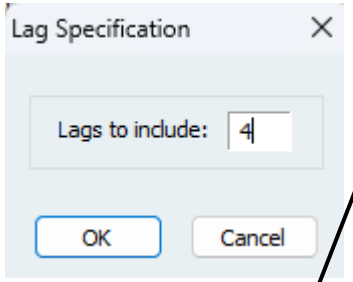
- If no autocorrelation in residuals, then regression coefficients are insignificant:

  - $\mathrm{H}_0: \psi_1 = 0,\ \psi_2 = 0, \ldots, \psi_m = 0$

Testing residuals for autocorrelation. LM test in EViews

Consider the estimated FDL model (2) for Poland inflation





Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 4 lags

F-statistic	6.999885	Prob. F(4,90)	0.0001
Obs*R-squared	23.72852	Prob. Chi-Square(4)	0.0001

Test Equation:
Dependent Variable: RESID
Method: Least Squares
Date: 02/04/26 Time: 16:57
Sample: 2000Q3 2025Q2
Included observations: 100
Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.049043	0.073155	0.670398	0.5043
DLOG(EU_CPI_OBS)*100	-0.127375	0.115463	-1.103163	0.2729
DLOG(EU_CPI_OBS(-1))*100	0.033313	0.115283	0.288963	0.7733
DLOG(S_EUR_OBS(-1))*100	-0.002155	0.014187	-0.151867	0.8796
GDP_GAP(-2)	0.013098	0.028800	0.454797	0.6504
RER_GAP(-1)	0.005326	0.010182	0.523011	0.6023
RESID(-1)	0.400971	0.106966	3.748583	0.0003
RESID(-2)	-0.021578	0.111084	-0.194251	0.8464
RESID(-3)	0.266456	0.112624	2.365892	0.0201
RESID(-4)	-0.036335	0.117191	-0.310051	0.7572

R-squared	0.237285	Mean dependent var	-3.61E-17
Adjusted R-squared	0.161014	S.D. dependent var	0.482603
S.E. of regression	0.442046	Akaike info criterion	1.299834
Sum squared resid	17.58643	Schwarz criterion	1.560351
Log likelihood	-54.99172	Hannan-Quinn criter.	1.405270
F-statistic	3.111060	Durbin-Watson stat	1.934566
Prob(F-statistic)	0.002661		

LM-statistic

p-value = 0.0001 < 0.05

We reject H_0: *no autocorrelation.*

Heteroskedasticity.
Q test on squared residuals

- Regression model: $y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t$

- Residuals: $\hat{u}_t = y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t} - \hat{\beta}_2 x_{2t} - \cdots - \hat{\beta}_k x_{kt}$

- Compute autocorrelations of the *squared* residuals rather than the residuals themselves.

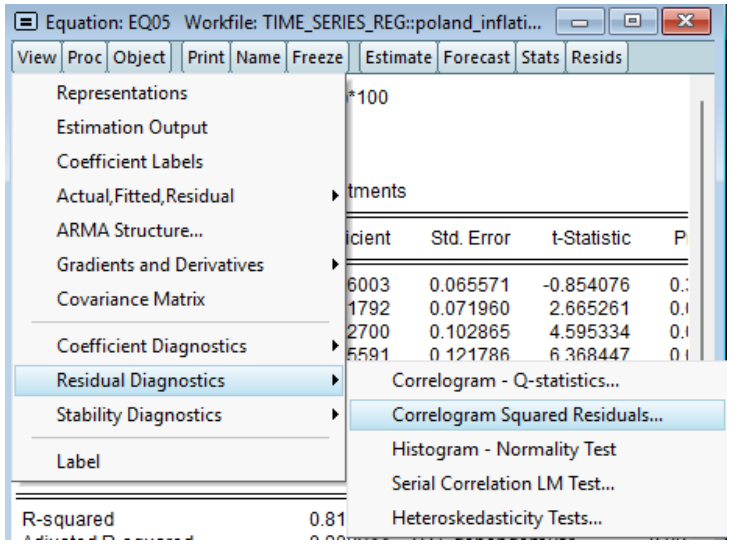
- If the error variance changes over time in a persistent way, then large squared residuals cluster together, and that clustering is autocorrelation in $\hat{u}_t^2$.

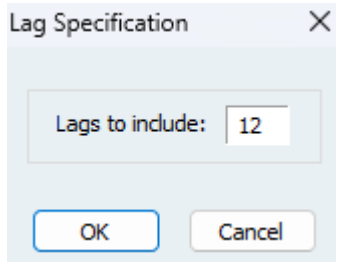
    - H_0: no heteroskedasticity up to order *m* (autocorrelation coefficients are jointly zero).

Heteroskedasticity.
Q test on squared residuals in EViews

Consider the estimated model for

Poland inflation





Date: 02/03/26 Time: 14:26
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.100	-0.100	1.0285	0.311
		2	0.038	0.029	1.1823	0.554
		3	-0.134	-0.129	3.0678	0.381
		4	0.052	0.026	3.3506	0.501
		5	-0.074	-0.061	3.9333	0.559
		6	-0.032	-0.064	4.0429	0.671
		7	-0.025	-0.023	4.1133	0.767
		8	0.055	0.035	4.4495	0.814
		9	0.005	0.006	4.4526	0.879
		10	0.031	0.025	4.5620	0.918
		11	-0.020	-0.009	4.6079	0.949
		12	0.107	0.101	5.9262	0.920

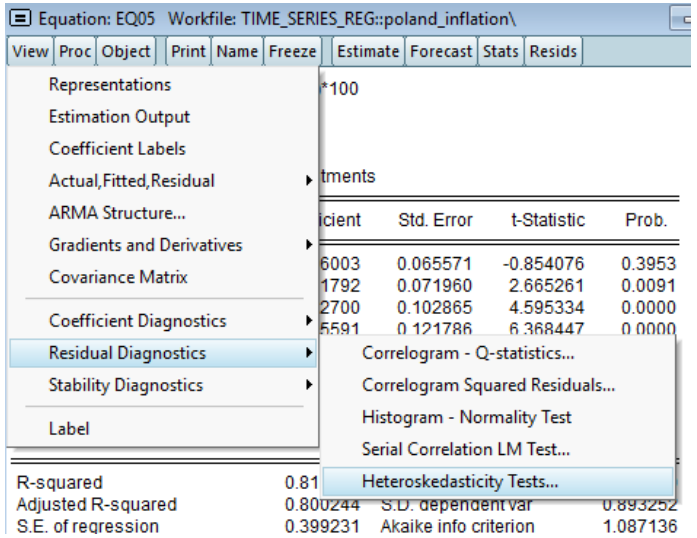
p-values > 0.05 for all lags

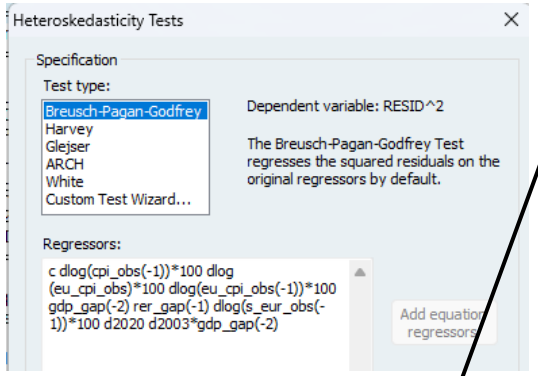
We fail to reject H_0: $homoskedasticity$.

Heteroskedasticity.
The Breusch-Pagan-Godfrey LM test

Consider the estimated model for

Poland inflation





Heteroskedasticity Test: Breusch-Pagan-Godfrey
Null hypothesis: Homoskedasticity

F-statistic	1.341690	Prob. F(8,91)	0.2333
Obs*R-squared	10.55062	Prob. Chi-Square(8)	0.2285
Scaled explained SS	7.384290	Prob. Chi-Square(8)	0.4958

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 02/03/26 Time: 13:51
Sample: 2000Q3 2025Q2
Included observations: 100

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.130241	0.030707	4.241424	0.0001
DLOG(CPI_OBS(-1))*100	0.053197	0.033699	1.578597	0.1179
DLOG(EU_CPI_OBS)*100	-0.011036	0.048172	-0.229096	0.8193
DLOG(EU_CPI_OBS(-1))*100	-0.032125	0.057032	-0.563271	0.5746
GDP_GAP(-2)	-7.36E-05	0.012932	-0.005688	0.9955
RER_GAP(-1)	-0.011508	0.004238	-2.715486	0.0079
DLOG(S_EUR_OBS(-1))*100	0.005857	0.006203	0.944229	0.3476
D2020	-0.160183	0.103249	-1.551422	0.1243
D2003*GDP_GAP(-2)	-0.026455	0.047522	-0.556701	0.5791

R-squared	0.105506	Mean dependent var	0.145041
Adjusted R-squared	0.026869	S.D. dependent var	0.189523
S.E. of regression	0.186959	Akaike info criterion	-0.430164
Sum squared resid	3.180788	Schwarz criterion	-0.195699
Log likelihood	30.50821	Hannan-Quinn criter.	-0.335272
F-statistic	1.341690	Durbin-Watson stat	2.392928
Prob(F-statistic)	0.233277		

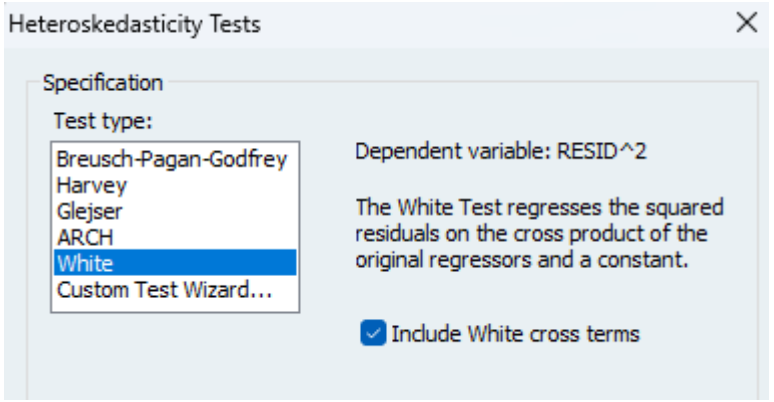
LM-statistic

p-value = 0.2285 > 0.05

We fail to reject H_0: $homoskedasticity$.

Heteroskedasticity. The White test

Consider the estimated model for Poland inflation



LM-statistic

p-value = 0.0925 > 0.05

We fail to reject H_0: *homoskedasticity*.

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	1.585954	Prob. F(34,65)	0.0553
Obs*R-squared	45.34252	Prob. Chi-Square(34)	0.0925
Scaled explained SS	31.73485	Prob. Chi-Square(34)	0.5791

Test Equation:
Dependent Variable: RESID^2
Method: Least Squares
Date: 02/03/26 Time: 13:54
Sample: 2000Q3 2025Q2
Included observations: 100
Collinear test regressors dropped from specification

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.071016	0.058746	1.208868	0.2311
DLOG(CPI_OBS(-1))*100^2	0.000196	0.000830	0.236103	0.8141
DLOG(CPI_OBS(-1))*100*DLOG(EU_CPI...	0.023680	0.098604	0.240152	0.8110
DLOG(CPI_OBS(-1))*100*DLOG(EU_CPI...	-0.000108	0.068960	-0.001563	0.9988
DLOG(CPI_OBS(-1))*100*GDP_GAP(-2)	-0.021249	0.037508	-0.566529	0.5730
DLOG(CPI_OBS(-1))*100*RER_GAP(-1)	0.001376	0.010643	0.129325	0.8975
DLOG(CPI_OBS(-1))*100*DLOG(S_EUR...	-0.016901	0.021479	-0.786858	0.4342
DLOG(CPI_OBS(-1))*100*D2020	-0.218581	0.305733	-0.714942	0.4772
DLOG(CPI_OBS(-1))*100*D2003*GDP_G...	0.343802	0.136913	2.511096	0.0145
DLOG(EU_CPI_OBS)*100^2	0.000982	0.001074	0.914523	0.3638
DLOG(EU_CPI_OBS)*100*DLOG(EU_CP...	-0.061964	0.125285	-0.494587	0.6226
DLOG(EU_CPI_OBS)*100*GDP_GAP(-2)	-0.020498	0.054084	-0.378999	0.7059
DLOG(EU_CPI_OBS)*100*RER_GAP(-1)	0.028673	0.019140	1.498076	0.1390
DLOG(EU_CPI_OBS)*100*DLOG(S_EUR...	-0.000411	0.037420	-0.010973	0.9913
DLOG(EU_CPI_OBS)*100*D2020	0.595684	1.816493	0.327931	0.7440
DLOG(EU_CPI_OBS)*100*D2003*GDP_...	0.670765	0.357623	1.875622	0.0652
DLOG(EU_CPI_OBS(-1))*100^2	-0.000653	0.001186	-0.551166	0.5834
DLOG(EU_CPI_OBS(-1))*100*GDP_GAP(-2)	0.021150	0.061443	0.344225	0.7318
DLOG(EU_CPI_OBS(-1))*100*RER_GAP(-1)	0.018363	0.022145	0.829224	0.4100
DLOG(EU_CPI_OBS(-1))*100*DLOG(S_...	-0.035683	0.023698	-1.505726	0.1370
DLOG(EU_CPI_OBS(-1))*100*D2020	0.165889	1.247541	0.132973	0.8946
DLOG(EU_CPI_OBS(-1))*100*D2003*GD...	-0.241496	0.298491	-0.809057	0.4214
GDP_GAP(-2)^2	0.005231	0.010435	0.501261	0.6179
GDP_GAP(-2)*RER_GAP(-1)	0.008249	0.006633	1.243747	0.2181
GDP_GAP(-2)*DLOG(S_EUR_OBS(-1))*100	0.006292	0.010094	0.623384	0.5352
GDP_GAP(-2)*D2020	0.104185	0.135987	0.766139	0.4464
GDP_GAP(-2)*D2003*GDP_GAP(-2)	-0.194009	0.076501	-2.536038	0.0136
GDP_GAP(-2)	0.003325	0.037635	0.088353	0.9299
RER_GAP(-1)^2	0.001088	0.000622	1.748809	0.0850
RER_GAP(-1)*DLOG(S_EUR_OBS(-1))*100	0.000851	0.001703	0.499702	0.6190
RER_GAP(-1)*D2003*GDP_GAP(-2)	-0.029848	0.012758	-2.339486	0.0224
RER_GAP(-1)	-0.038437	0.015267	-2.517625	0.0143
DLOG(S_EUR_OBS(-1))*100^2	0.000163	0.000177	0.922564	0.3596
DLOG(S_EUR_OBS(-1))*100*D2003*GD...	0.019540	0.027940	0.699360	0.4868
D2003*GDP_GAP(-2)	-0.598186	0.358704	-1.667632	0.1002

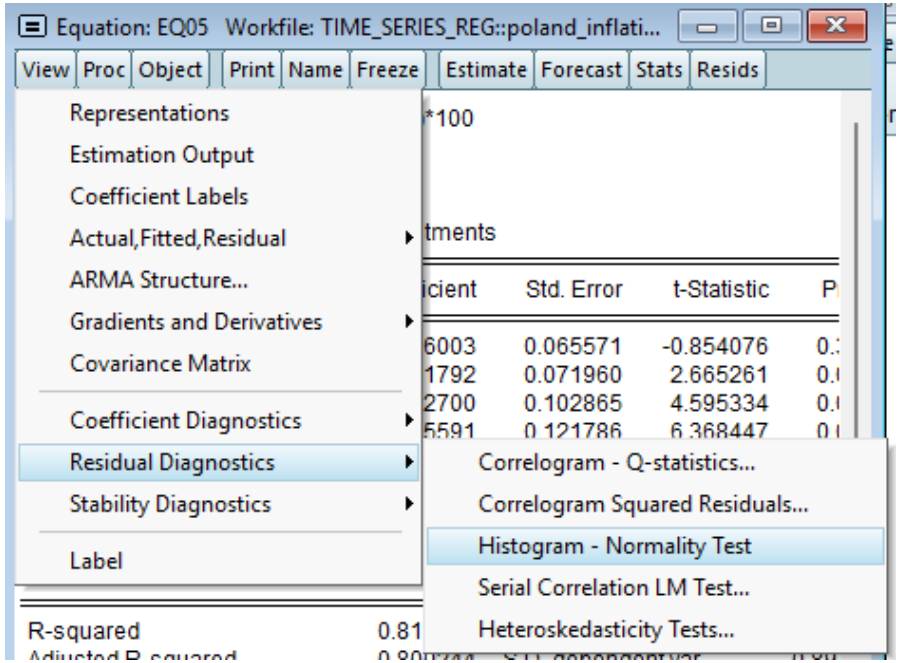
R-squared	0.453425	Mean dependent var	0.145041
Adjusted R-squared	0.167525	S.D. dependent var	0.189523
S.E. of regression	0.172921	Akaike info criterion	-0.402751
Sum squared resid	1.943600	Schwarz criterion	0.509059
Log likelihood	55.13755	Hannan-Quinn criter.	-0.033725
F-statistic	1.585954	Durbin-Watson stat	2.635682
Prob(F-statistic)	0.055345		

Test for normality

We can use the **Jarque-Bera statistic** to test the normality of the residuals.

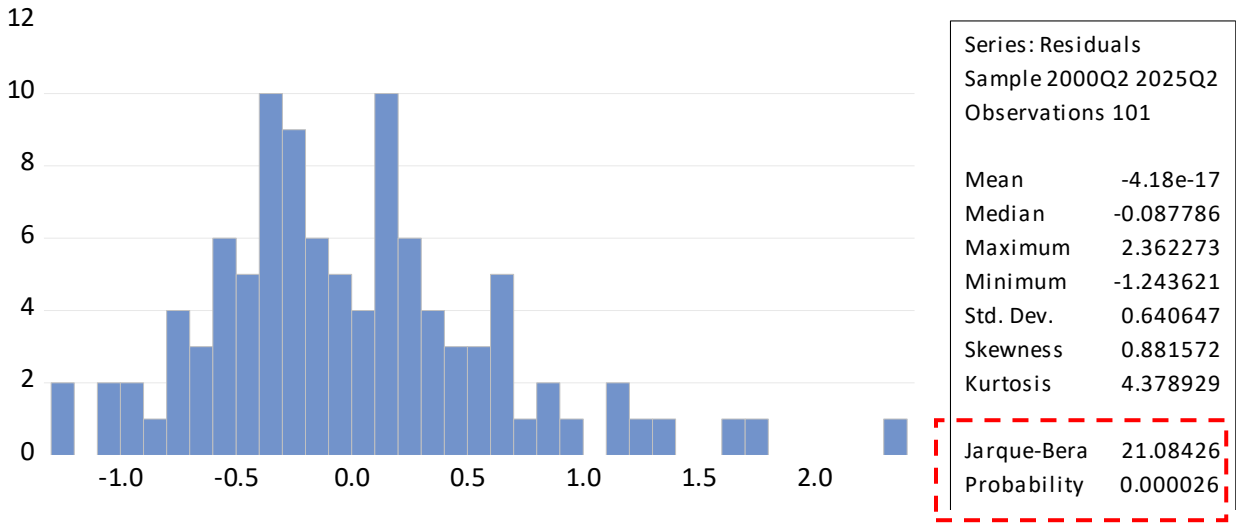
If the *p*-value of the Jarque-Bera statistic is less than 0.05 (for a 5% significance level), then we reject the null hypothesis of normality.

Testing residuals normality



Test for normality. Example in EViews

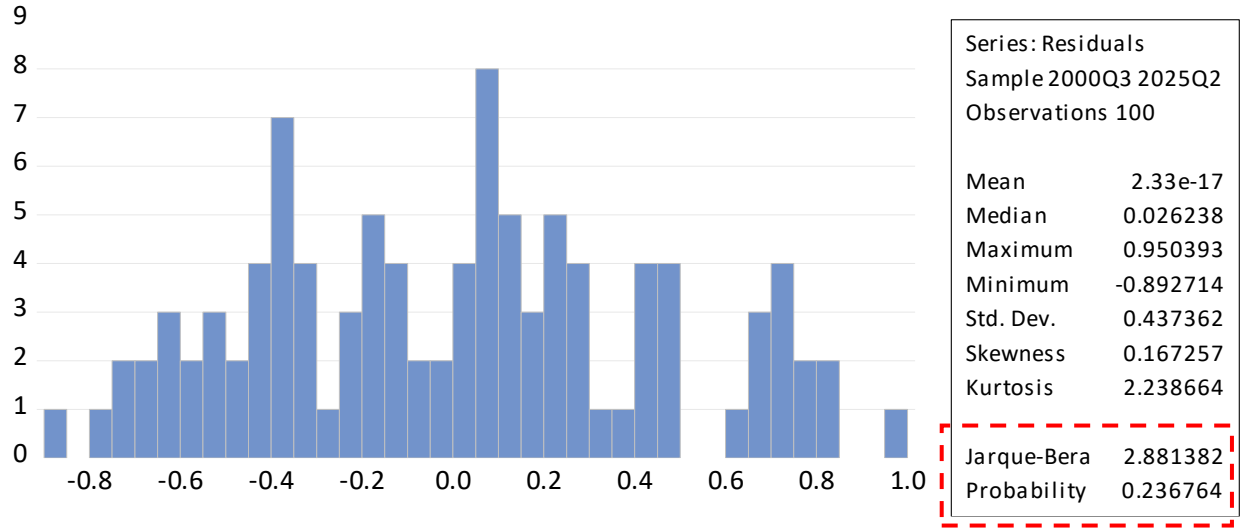
Poland inflation model 1 (static)



p-value = 0.00 < 0.05

We reject H_0: *the residuals are normally distributed.*

Poland inflation model 3 (ADL)



p-value = 0.237 > 0.05

We fail to reject H_0: *the residuals are normally distributed.*

6. Dealing with error irregularities

Residual autocorrelation

- Two cases:
  - Case 1, strictly exogenous regressors.
    - Example: AR(1) u_t

$$y_t = \beta z_t + u_t \qquad\qquad u_t = \rho u_{t-1} + v_t$$

  - Case 2, a lagged dependent variable in the equation.
    - Example: y_{t-1} and AR(1) u_t

$$y_t = \beta_1 y_{t-1} + \beta_2 z_t + u_t \qquad\qquad u_t = \rho u_{t-1} + v_t$$

Dealing with autocorrelation.
Case 1: No lagged dependent variable

- In general, add lags y_{t-1}, z_{t-1}, … :

$$y_t = \beta z_t + u_t \quad \Rightarrow \quad y_t = \beta_1 z_t + \beta_2 y_{t-1} + \beta_3 z_{t-1} + u_t$$

Poland inflation model 2 (FDL)

Date: 02/07/26 Time: 14:38
Sample (adjusted): 2000Q3 2025Q2
Q-statistic probabilities adjusted for 5 dynamic regressors

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.423	0.423	18.405	0.000
		2	0.229	0.062	23.883	0.000
		3	0.293	0.215	32.932	0.000
		4	0.149	-0.063	35.277	0.000
		5	0.069	-0.014	35.790	0.000
		6	0.170	0.123	38.942	0.000
		7	0.088	-0.041	39.793	0.000
		8	-0.092	-0.168	40.732	0.000
		9	0.063	0.132	41.179	0.000
		10	0.027	-0.046	41.265	0.000
		11	-0.079	-0.040	41.979	0.000
		12	-0.011	-0.001	41.993	0.000

*Probabilities may not be valid for this equation specification.

Poland inflation model 3 (ADL)

Date: 02/07/26 Time: 14:38
Sample (adjusted): 2000Q3 2025Q2
Q-statistic probabilities adjusted for 6 dynamic regressors

Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob*
		1	0.046	0.046	0.2164	0.642
		2	0.025	0.023	0.2809	0.869
		3	0.259	0.258	7.3548	0.061
		4	0.089	0.072	8.1946	0.085
		5	-0.017	-0.033	8.2250	0.144
		6	0.076	0.009	8.8574	0.182
		7	0.114	0.077	10.273	0.174
		8	-0.206	-0.223	14.956	0.060
		9	0.058	0.057	15.334	0.082
		10	0.004	-0.052	15.335	0.120
		11	-0.098	-0.007	16.432	0.126
		12	-0.045	-0.041	16.671	0.162

*Probabilities may not be valid for this equation specification.

Dealing with autocorrelation.
Case 1: No lagged dependent variable (cont…)

- Adding AR(p) will correct the autocorrelation in the residuals. But the long-run multipliers and interpretations will be different.

Poland inflation model 3 (ADL)

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:15
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000209	0.071828	0.002911	0.9977
DLOG(CPI_OBS(-1))*100	0.337564	0.075042	4.498336	0.0000
DLOG(EU_CPI_OBS)*100	0.415841	0.114470	3.632762	0.0005
DLOG(EU_CPI_OBS(-1))*100	0.562013	0.129821	4.329120	0.0000
DLOG(S_EUR_OBS(-1))*100	0.045946	0.014305	3.211898	0.0018
GDP_GAP(-2)	0.059109	0.030396	1.944599	0.0548
RER_GAP(-1)	0.020443	0.010216	2.001096	0.0483

R-squared	0.760264	Mean dependent var	0.791340
Adjusted R-squared	0.744797	S.D. dependent var	0.893252
S.E. of regression	0.451250	Akaike info criterion	1.313839
Sum squared resid	18.93728	Schwarz criterion	1.496201
Log likelihood	-58.69197	Hannan-Quinn criter.	1.387645
F-statistic	49.15433	Durbin-Watson stat	1.868432
Prob(F-statistic)	0.000000		

Persistence is **explicitly modeled – structural persistence** (sticky prices)

Poland inflation model 3 with AR(1)

Dependent Variable: DLOG(CPI_OBS)*100
Method: ARMA Maximum Likelihood (OPG - BHHH)
Date: 02/06/26 Time: 16:27
Sample: 2000Q3 2025Q2
Included observations: 100
Convergence achieved after 10 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.129712	0.111804	1.160175	0.2490
DLOG(EU_CPI_OBS)*100	0.446887	0.099400	4.495867	0.0000
DLOG(EU_CPI_OBS(-1))*100	0.813425	0.083110	9.787353	0.0000
DLOG(S_EUR_OBS(-1))*100	0.026768	0.015684	1.706679	0.0913
GDP_GAP(-2)	0.078635	0.024561	3.201597	0.0019
RER_GAP(-1)	0.038419	0.014426	2.663237	0.0091
AR(1)	0.478962	0.109363	4.379574	0.0000
SIGMASQ	0.183759	0.032162	5.713500	0.0000

R-squared	0.767370	Mean dependent var	0.791340
Adjusted R-squared	0.749670	S.D. dependent var	0.893252
S.E. of regression	0.446921	Akaike info criterion	1.306355
Sum squared resid	18.37594	Schwarz criterion	1.514768
Log likelihood	-57.31774	Hannan-Quinn criter.	1.390703
F-statistic	43.35394	Durbin-Watson stat	2.041281
Prob(F-statistic)	0.000000		

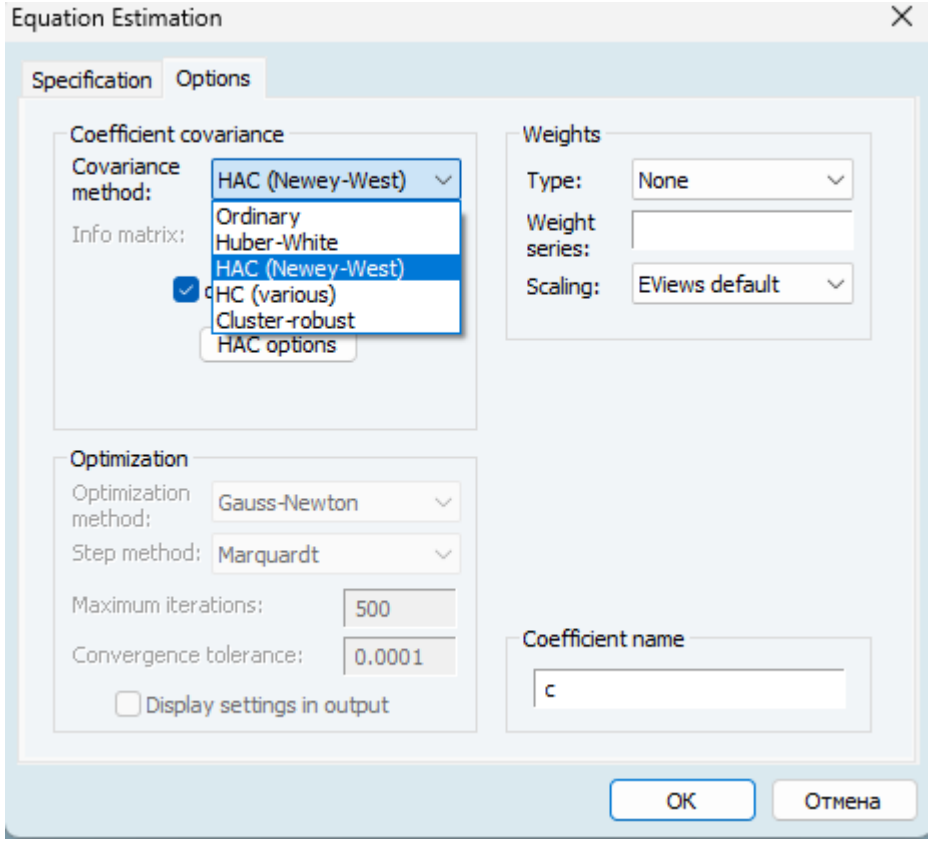
Inverted AR Roots	.48

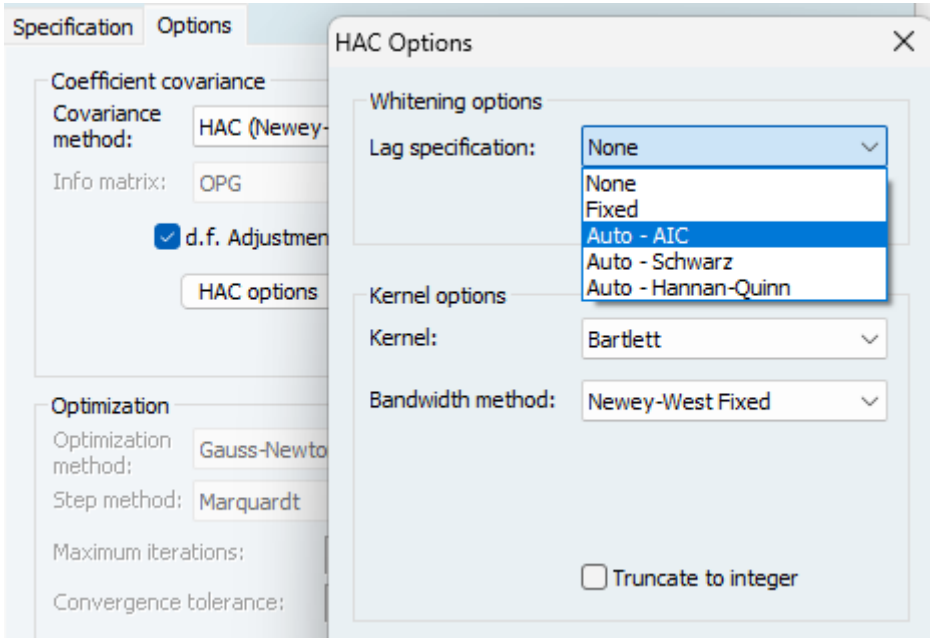
Persistence is **pushed into the error term –** shocks are persistent, but no built-in inflation inertia

This is ρ in
$u_t = \rho u_{t-1} + v_t$

Dealing with autocorrelation.
Case 1: No lagged dependent variable (cont…)

- If the aim is to check significance of a regression: Robust standard errors





Dealing with autocorrelation.
Case 1: No lagged dependent variable (cont…)

- If the aim is to check significance of a regression: Robust standard errors

Poland inflation model 2 (FDL)

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 01/24/26 Time: 16:09
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.076033	0.076634	0.992159	0.3237
DLOG(EU_CPI_OBS)*100	0.526008	0.122728	4.285982	0.0000
DLOG(EU_CPI_OBS(-1))*100	0.822268	0.127554	6.446423	0.0000
DLOG(S_EUR_OBS(-1))*100	0.040880	0.015652	2.611813	0.0105
GDP_GAP(-2)	0.103230	0.031577	3.269132	0.0015
RER_GAP(-1)	0.029243	0.011005	2.657226	0.0093

R-squared	0.708101	Mean dependent var	0.791340
Adjusted R-squared	0.692575	S.D. dependent var	0.893252
S.E. of regression	0.495272	Akaike info criterion	1.490705
Sum squared resid	23.05767	Schwarz criterion	1.647016
Log likelihood	-68.53527	Hannan-Quinn criter.	1.553967
F-statistic	45.60594	Durbin-Watson stat	1.112405
Prob(F-statistic)	0.000000		

Poland inflation model 2 (FDL) with HAC *s.e.*

Dependent Variable: DLOG(CPI_OBS)*100
Method: Least Squares
Date: 02/08/26 Time: 14:27
Sample (adjusted): 2000Q3 2025Q2
Included observations: 100 after adjustments
HAC standard errors & covariance (Prewhitening with lags = 1 from AIC
maxlags = 4, Bartlett kernel, Newey-West fixed bandwidth = 4.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.076033	0.107368	0.708154	0.4806
DLOG(EU_CPI_OBS)*100	0.526008	0.114070	4.611290	0.0000
DLOG(EU_CPI_OBS(-1))*100	0.822268	0.145879	5.636653	0.0000
DLOG(S_EUR_OBS(-1))*100	0.040880	0.021952	1.862194	0.0657
GDP_GAP(-2)	0.103230	0.057688	1.789446	0.0768
RER_GAP(-1)	0.029243	0.012376	2.362815	0.0202

R-squared	0.708101	Mean dependent var	0.791340
Adjusted R-squared	0.692575	S.D. dependent var	0.893252
S.E. of regression	0.495272	Akaike info criterion	1.490705
Sum squared resid	23.05767	Schwarz criterion	1.647016
Log likelihood	-68.53527	Hannan-Quinn criter.	1.553967
F-statistic	45.60594	Durbin-Watson stat	1.112405
Prob(F-statistic)	0.000000	Wald F-statistic	47.15719
Prob(Wald F-statistic)	0.000000		

Dealing with autocorrelation.
Case 2: substitute the error process into the equation

- A model includes lagged dependent variable y_{t-1}, and u_t are autocorrelated:

  - Example: y_{t-1} and AR(1) u_t

$$y_t = \beta_1 y_{t-1} + \beta_2 z_t + u_t \qquad\qquad u_t = \rho u_{t-1} + v_t$$

- Add lags: in general – $y_{t-2},\ z_{t-1}$, …

$$y_t = \beta_1 y_{t-1} + \beta_2 z_t + u_t \quad \Rightarrow \quad y_t = \alpha_1 y_{t-1} + \alpha_2 z_t + \alpha_3 y_{t-2} + \alpha_4 z_{t-1} + v_t$$

Heteroskedastic errors

- Heteroskedasticity:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \cdots + \beta_k x_{kt} + u_t \qquad Var(u_t) = \sigma_t^2$$

- OLS remains unbiased and consistent.

- OLS is no longer the minimum-variance linear unbiased estimator, and the conventional standard errors are wrong

- For forecasting: the mean forecast is unaffected. What breaks is the interval.

Dealing with heteroskedasticity

- **Model the variance.** Specify a conditional-variance equation and estimate jointly by maximum likelihood, ARCH or GARCH being the usual choice.

- **Correct the standard errors.** Use White's estimator for heteroskedasticity alone, Newey–West (HAC) when the errors are also autocorrelated. Do this when the variance is a nuisance and the goal is inference on coefficients.

Dealing with heteroskedasticity. Example in EViews

Belarus inflation model

Dependent Variable: DLOG(CPI_BY)*100
Method: Least Squares
Date: 02/11/26 Time: 14:24
Sample (adjusted): 2003Q4 2025Q4
Included observations: 89 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(CPI_BY(-1))*100	0.239701	0.033938	7.062931	0.0000
GDP_GAP	0.316023	0.064825	4.874994	0.0000
DLOG(NER)*100	0.534344	0.023143	23.08852	0.0000
DLOG(CPI_RU(-2))*100	0.227922	0.101554	2.244347	0.0275
C	0.625518	0.253518	2.467356	0.0157
D2015*DLOG(NER)*100	-0.352716	0.055143	-6.396402	0.0000
D2022_Q1	5.981305	1.226537	4.876580	0.0000
D2014_Q4	5.264530	1.263546	4.166473	0.0001

R-squared	0.934066	Mean dependent var	2.948146
Adjusted R-squared	0.928368	S.D. dependent var	4.515005
S.E. of regression	1.208403	Akaike info criterion	3.302064
Sum squared resid	118.2792	Schwarz criterion	3.525761
Log likelihood	-138.9418	Hannan-Quinn criter.	3.392230
F-statistic	163.9288	Durbin-Watson stat	1.588496
Prob(F-statistic)	0.000000		

Residuals are autocorrelated

Breusch-Godfrey Serial Correlation LM Test:
Null hypothesis: No serial correlation at up to 4 lags

F-statistic	4.446216	Prob. F(4,77)	0.0028
Obs*R-squared	16.69943	Prob. Chi-Square(4)	0.0022

… and heteroskedastic

Heteroskedasticity Test: White
Null hypothesis: Homoskedasticity

F-statistic	4.659193	Prob. F(18,70)	0.0000
Obs*R-squared	48.51009	Prob. Chi-Square(18)	0.0001
Scaled explained SS	67.42056	Prob. Chi-Square(18)	0.0000



Dependent Variable: DLOG(CPI_BY)*100
Method: Least Squares
Date: 02/11/26 Time: 14:28
Sample (adjusted): 2003Q4 2025Q4
Included observations: 89 after adjustments
HAC standard errors & covariance (Bartlett kernel, Newey-West fixed bandwidth = 4.0000)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DLOG(CPI_BY(-1))*100	0.239701	0.056542	4.239307	0.0001
GDP_GAP	0.316023	0.115173	2.743898	0.0075
DLOG(NER)*100	0.534344	0.062792	8.509804	0.0000
DLOG(CPI_RU(-2))*100	0.227922	0.084156	2.708319	0.0082
C	0.625518	0.269987	2.316842	0.0230
D2015*DLOG(NER)*100	-0.352716	0.092767	-3.802195	0.0003
D2022_Q1	5.981305	0.271597	22.02273	0.0000
D2014_Q4	5.264530	0.886768	5.936760	0.0000

R-squared	0.934066	Mean dependent var	2.948146
Adjusted R-squared	0.928368	S.D. dependent var	4.515005
S.E. of regression	1.208403	Akaike info criterion	3.302064
Sum squared resid	118.2792	Schwarz criterion	3.525761
Log likelihood	-138.9418	Hannan-Quinn criter.	3.392230
F-statistic	163.9288	Durbin-Watson stat	1.588496
Prob(F-statistic)	0.000000		

Summary

What breaks the coefficients

- Regressors correlated with the errors (TS.3).
- A lagged dependent variable together with autocorrelated errors.
- An unmodelled break. Parameters that average two regimes.

What breaks only the standard errors

- Autocorrelation with exogenous regressors.
- Heteroskedasticity.

What breaks almost nothing

- Non-normal errors in a reasonable sample. Inference is asymptotic rather than exact.

What to do

- Diagnose before remedying.
- Prefer respecification to robust standard errors when the model is for forecasting; robust standard errors are enough when it is for inference.
- Every remedy needs justification, not just an improved diagnostic.

Appendix.
HP Filter in EViews

Hodrick-Prescott (HP) filter

HP filter:

- minimizes the cyclical component of the time series (the output gap),
- while minimizing the changes in the trend (potential output).

To control for the smoothness of the trend **the parameter of λ** is used.

The smaller the value of λ, the more volatile the trend becomes.

The default setting for λ for data of quarterly frequency is 1600.

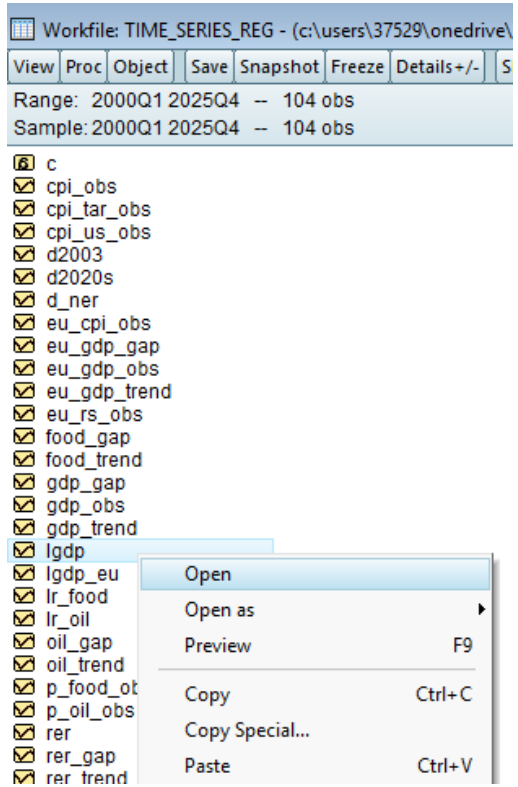
Typical value for annual data is 100, and for monthly data – 14400 or 129600.

Hodrick-Prescott (HP) filter in EViews (1)

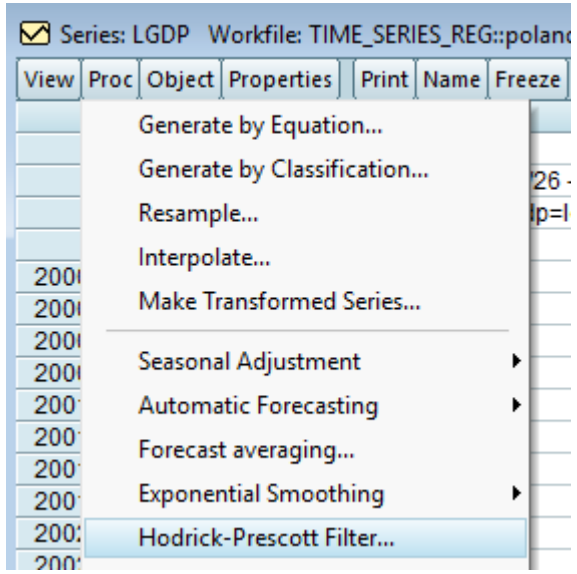
- If the variable is in levels (GDP, REER, etc.), take the logarithm (multiply by 100 for a percentage interpretation).

- If the variable is in percent (real interest rate, unemployment rate, etc.), no conversion is necessary.

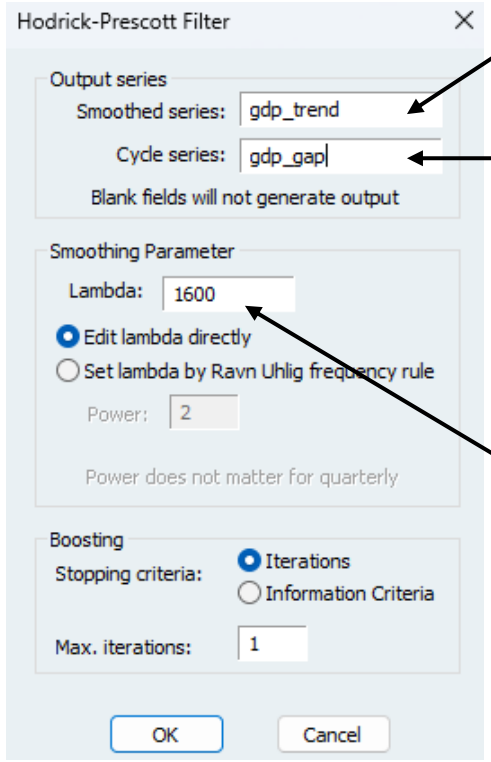
1. Open the series (after transformation if needed)



2. Proc / Hodrick-Prescott Filter



3. Filter settings

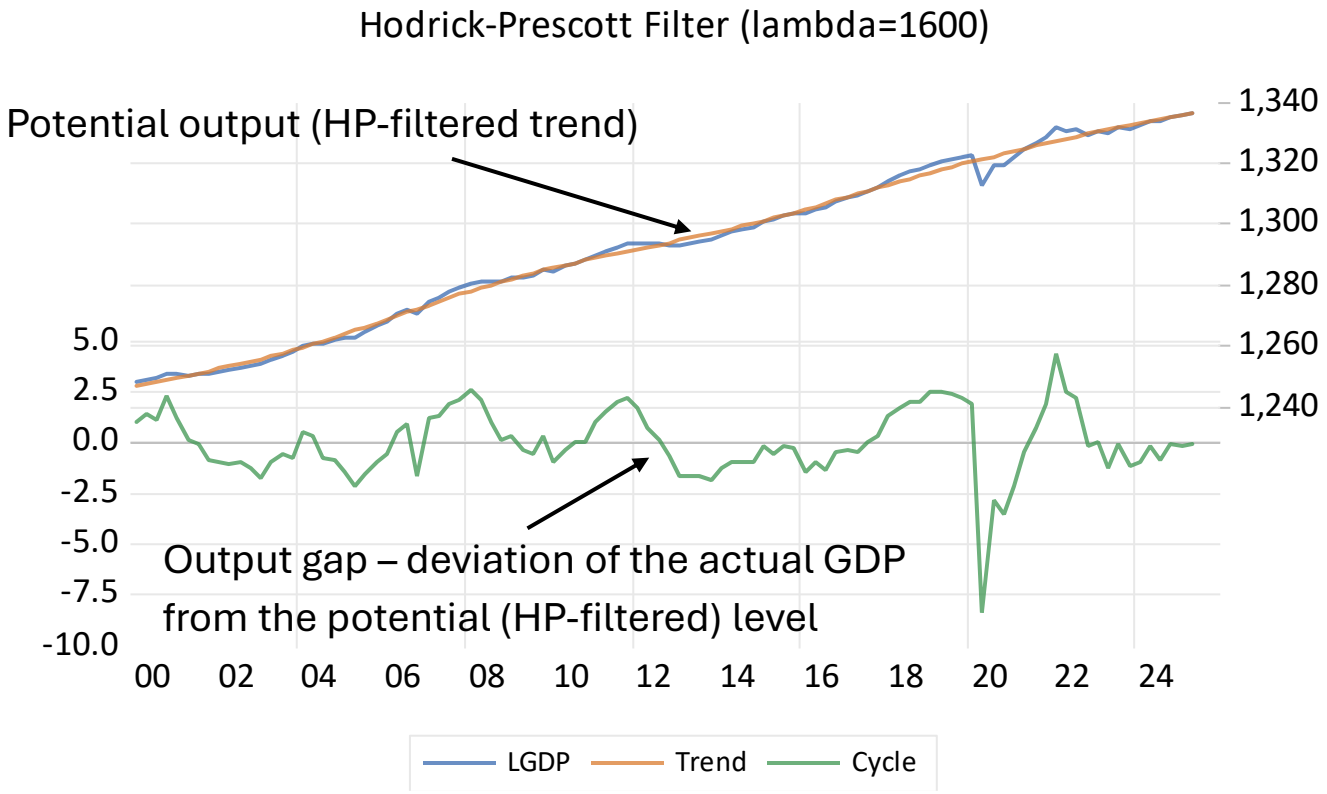


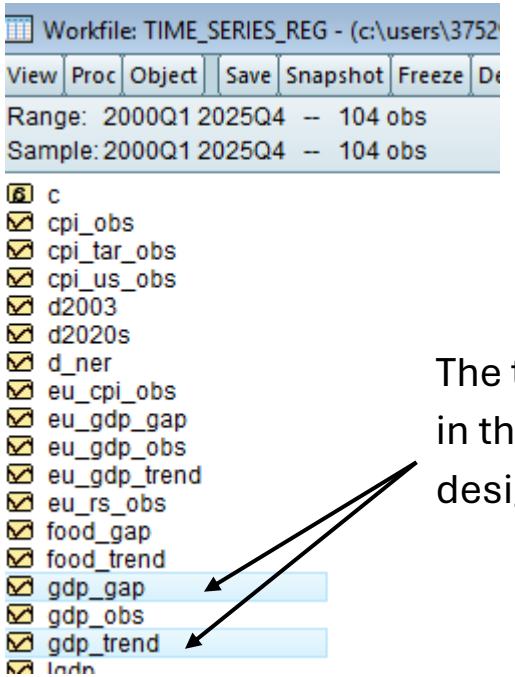
Specify the name for the trend series

Specify the name for the gap series

Use the default value, unless you have compelling reasons to change it (e.g., a business cycle duration that differs significantly from that of the US).

Hodrick-Prescott (HP) filter in EViews (2)





The trend and cycle are saved in the workfile under the designated names.